Household Vehicle Purchase Timing: Selected Papers and Literature Review

The Purchase Decision Within the Auto-Loan Research Exercise

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Discussion draft for Eddie

Abstract

This review covers the papers a model of household vehicle-purchase timing would build on, as the step that precedes auto-loan demand. Section 2 walks through each selected paper's mathematics in the authors' notation: the setup, the derivation step by step, the assumption each step uses, the results, and the strengths and limitations that follow. The papers cover dynamic discrete choice of when to buy; identification and estimation, including latent household types and which counterfactuals survive normalization; latent household states; liquidity in the purchase decision; and initial conditions. Three are pending full-text review. A compatibility summary states which components combine directly and which require added assumptions. Section 3 reviews the broader evidence through Need, Ability, and Choice. Section 4 lists further references identified for later reading. How the components enter our model is set out in a companion research proposal.

1 Purpose and Scope

The broader auto-loan exercise asks which existing bank customers may take an auto loan over a future six- or twelve-month interval. Under the scope revised on 16 October 2026, the immediate target is the household decision that precedes the loan: the probability that a household purchases a vehicle within the horizon. A vehicle purchase, borrowing to fund it, and originating a loan with a particular lender are different events; this review concerns the first. Creditworthiness, the financing choice, and household risk attitude are out of scope at this stage.

This document reviews the papers that a model of that probability would draw on. Section 2 walks through selected papers in the authors' own notation: the setup (what is a given state, what is chosen, and who observes what), the derivation step by step with the assumption each step uses, the results, the assumptions stated by the authors and those the argument also relies on, and the strengths and limitations that follow. The papers are grouped by the model component they supply: the dynamic choice of when to buy (C1), identification and estimation, including latent types and counterfactuals (C2), a latent household state (C3), liquidity in the purchase decision (C4), and initial conditions with a hazard baseline (C5). A compatibility summary states which components combine directly and which require added assumptions. Section 3 reviews the broader evidence on why households buy vehicles when they do, organized by *Need* (what creates pressure to change the household's vehicles), *Ability* (which transactions are financially feasible), and *Choice* (which feasible action, including waiting, is preferred).

How these components are used is set out in the companion research proposal: its formulation, which steps it keeps, and which it changes, citing each paper's walkthrough here by section and

step number. Where an entry checks or reconstructs a step, or illustrates it with invented values, the entry says so; such material is ours, not evidence from the paper.

2 Selected Papers: Equations, Assumptions, Strengths, and Limitations

Selection. A paper is included if it supplies an equation or identification result that a model of household purchase timing would use directly: the monthly choice probability (C1), identification and estimation, including latent types and counterfactuals (C2), a latent state with transition and measurement equations (C3), liquidity in the purchase decision (C4), or initial conditions and a hazard baseline (C5). Creditworthiness, financing choice, and household risk attitude are out of scope for this stage.

How to read each entry. Each paper is a walkthrough of its math: the setup (what is a given state, what is chosen, and who observes what), the derivation step by step in the paper's own notation and equation numbers with the assumption each step uses, the results that matter for us, the assumptions the authors state and those we think the argument also relies on, and, where one arises from the paper itself, a question for discussion. Entries stay in the authors' notation. Which steps our model keeps or changes is set out in the companion research proposal, which cites the steps numbered here. Three papers are pending full-text review and are marked **TODO:** pending.

2.1 C1. Dynamic choice of when to buy

2.1.1 Rust (1987): bus engine replacement

Why it is included: it is the core of a purchase-timing model, a monthly keep-or-replace decision whose choice probability is a logit in current utility plus discounted future value.

Setup. One decision maker (the maintenance manager) and 104 buses, monthly, December 1974 to May 1985 (p. 1001). *Given state:* accumulated mileage since the last replacement, x_t, observed by both agent and researcher; cost shocks $\epsilon_t = (\epsilon_t(0), \epsilon_t(1))$, observed only by the agent. *Action:* $i_t \in C(x_t) = \{0, 1\}$, keep or replace. *Objective:* minimize expected discounted cost.

Math walkthrough.

1. **Flow utility** (5.1), p. 1015:

$$u(x_t, i, \theta_1) + \epsilon_t(i) = \begin{cases} -RC - c(0, \theta_1) + \epsilon_t(1) & i = 1, \\ -c(x_t, \theta_1) + \epsilon_t(0) & i = 0. \end{cases}$$

   Assumptions used: shocks enter additively; operating cost c depends only on mileage; normalization $c(0, \theta_1) = 0$, because adding a constant to utility does not change choices (p. 1015).

2. **Transition** (5.2), p. 1015: $p(x_{t+1} \mid x_t, i, \theta_3) = g(x_{t+1} - x_t, \theta_3)$ if $i = 0$ and $g(x_{t+1} - 0, \theta_3)$ if $i = 1$. Replacement resets mileage, so the process "regenerates." *Assumptions used:* first-order Markov mileage; increments independent of the shocks; discretized into 90 cells of 5,000 miles.

3. **Dynamic program** (4.1)–(4.4), pp. 1010: $V_\theta(x_t, \epsilon_t) = \max_i [u(x_t, i, \theta_1) + \epsilon_t(i) + \beta E V_\theta(x_t, \epsilon_t, i)]$, with a stationary optimal policy $i_t = f(x_t, \epsilon_t, \theta)$ (4.3). *Assumptions used:* infinite horizon, stationarity, discounting with $\beta < 1$, and regularity conditions in Rust's companion paper.

4. **Conditional independence (CI)** (4.7), p. 1011: $p(x_{t+1}, \epsilon_{t+1} \mid x_t, \epsilon_t, i) = q(\epsilon_{t+1} \mid x_{t+1})\, p(x_{t+1} \mid x_t, i)$. *Consequence:* EV_θ depends on (x, i) only, not on ϵ. *Assumption used:* next period's shock depends on the past only through x_{t+1}, and x_{t+1} does not depend on ϵ_t. Rust calls this "a strong assumption" (p. 1011). This is the step that makes the model computable.

5. **Choice probability** (Theorem 1, (4.8)–(4.10), p. 1011): $P(i \mid x, \theta)$ is the derivative of the expected maximum ("social surplus") G with respect to $u(x, i)$, and EV_θ is the unique fixed point of a contraction. *Assumption used:* CI plus a known shock density q.

6. **Extreme-value case** (4.11)–(4.14), p. 1012. With i.i.d. type-I extreme-value shocks, $G = \log \sum_j \exp[u(x, j) + \beta EV(x, j)]$ and

$$P(i \mid x, \theta) = \frac{\exp\{u(x, i, \theta_1) + \beta EV_\theta(x, i)\}}{\sum_j \exp\{u(x, j, \theta_1) + \beta EV_\theta(x, j)\}},$$

$$EV_\theta(x, i) = \int_y \log\Big\{ \sum_{j \in C(y)} \exp[u(y, j, \theta_1) + \beta EV_\theta(y, j)] \Big\} p(dy \mid x, i, \theta_3).$$

   Assumptions used: EV1 distribution, independent across alternatives and time; mean and variance normalized (pp. 1015–1016). *Check:* if $p(\cdot \mid x, i)$ does not depend on i, then $EV(x, 0) = EV(x, 1)$ cancels and the static logit results (p. 1012). *Note:* as printed, the location in (4.11) is $+\gamma$, which conflicts with the mean-zero normalization on p. 1015; it changes only the level of EV, not P.

7. **Likelihood** (Theorem 2, (4.15), p. 1013):

$$\ell(x_1, \ldots, x_T, i_1, \ldots, i_T \mid x_0, i_0, \theta) = \prod_{t=1}^{T} P(i_t \mid x_t, \theta)\, p(x_t \mid x_{t-1}, i_{t-1}, \theta_3).$$

   Past choices affect current choices only through x_t ("structural state dependence," p. 1013). *Assumptions used:* CI; buses independent of one another (fn. 11); the initial state (x_0, i_0) is taken as given.

8. **Estimation** (pp. 1016–1019): stage 1 estimates θ_3 from the transition factor alone; stage 2 estimates (θ_1, RC) from the choice factor, solving the fixed point (4.14) at every trial parameter (nested fixed point); stage 3 runs full maximum likelihood for efficient standard errors.

9. **Identification** (p. 1023, fn. 12; Table VIII, p. 1020). The cost level and the shock location and scale are normalized. β is collinear with RC (both delay replacement), so it is fixed at .9999 or 0. With a nonparametric cost function the log-likelihood is identical at both values (-110.832 for groups 1–3), so β is not identified without restricting u.

Results to note. $\beta = .9999$ fits better than $\beta = 0$ in the pooled sample (LR 12.782, $p = .0035$) and in groups 1–3 ($p = .029$), not significantly in group 4 ($p = .053$) (Table IX). Conditional independence is rejected for groups 1–3 ($p = .0043$) but not group 4 (Table XI).

Assumptions: stated and implicit.

- *Stated by the author:* additive i.i.d. EV1 shocks; CI; Markov mileage; infinite-horizon stationary problem; β fixed; independence across buses.
- *Implicit, in our reading:* no persistent unobserved differences within a bus group (the CI rejection suggests otherwise); the agent knows the transition law; replacement is always affordable (no budget constraint); one asset per decision; behavior stable over ten years.

2.1.2 Schiraldi (2011): automobile replacement with transaction costs

Why it is included: it is the car-specific version of Rust's keep-or-replace engine, with a resale market, a per-purchase transaction cost τ_{jt}, and a purchase probability that collapses the vehicle menu into one dynamic inclusive value.

Setup. Aggregate registration shares for the Province of Isernia, Italy, 1994–2004, by product (age, engine size, fuel, manufacturer group; 2,178 product-years, p. 268). Annual periods, $\beta = 0.9$ fixed (p. 276). No household choices are observed: "consumer i" is a simulated type, 80 Halton draws of income y_i and age taste υ_i (p. 279). *Given state:* held car $k \in J_{t-1} \cup \{0\}$ and market information Ω_{it} (attributes, prices, availability), seen by consumer and researcher; ξ_{jt}, τ_{jt}, and shocks ϵ_{ijt}, seen by the consumer only. *Actions:* keep k, sell and buy $j \in J_t$ (new or used), or sell and hold no car ($j = 0$, $p_{0t} = 0$). At most one car per consumer (p. 272).

Math walkthrough.

1. **Flow utilities** (1)–(3), pp. 272–273: keep $\tilde{u}^k_{it} = x_{kt}\alpha^x_i + \xi_{kt} + \epsilon_{ikt}$; buy $u^{kj}_{it} = x_{jt}\alpha^x_i + \xi_{jt} - \alpha^p_i p_{jt} - \tau_{jt} + \alpha^p_i p_{kt} + \epsilon_{ijt}$; exit $u^{k0}_{it} = \alpha^p_i p_{kt} + \epsilon_{i0t}$. Heterogeneity: $\alpha^p_i = \alpha^p / y_i$, $\alpha^{age}_i = \alpha^{age} + \upsilon_i \sigma_{\alpha^{age}}$ (pp. 276–277). *Assumptions used:* only the buyer pays τ; the seller receives p_{kt} in full; full information, no adverse selection (p. 272).

2. **Bellman equation** (4), p. 273: $\hat{V}_i(k, \epsilon_{it}, \Omega_{it})$ is the maximum of the keep branch, the best buy branch $\max_j u^{kj}_{it} + \beta \mathbb{E}[\hat{V}_i(j, \cdot) \mid \Omega_{it}, \epsilon_{it}]$, and $1[k \neq 0]$ times the exit branch. Its state is too large to solve (p. 273).

3. **Price shift and rental price** (5)–(6), pp. 273–274. Set $V_i(k, \cdot) \equiv \hat{V}_i(k, \cdot) - \alpha^p_i p_{kt}$ and

$$\phi_{ijt} \equiv x_{jt}\alpha^x_i + \xi_{jt} - \alpha^p_i (p_{jt} - \beta \mathbb{E}_t[p_{jt+1}]), \qquad \phi_{i0t} = 0.$$

   The bracket is a one-year rental price. The shift cancels the resale term in (2)–(3); in our rewriting, the branches of (5) become exit $\epsilon_{i0t} + \beta \mathbb{E}[V_i(0)]$, buy $\phi_{ijt} - \tau_{jt} + \epsilon_{ijt} + \beta \mathbb{E}[V_i(j)]$, keep $1[k \neq 0](\phi_{ikt} + \epsilon_{ikt} + \beta \mathbb{E}[V_i(k)])$. *Assumptions used:* no selling cost; prices are in Ω_{it}.

4. **Inclusive value** (7), p. 275, with $EV_i(j, \Omega_{it}) = \int V_i(j, \epsilon_{it}, \Omega_{it})\, dP_\epsilon$ (p. 274):

$$\delta_{it} = \ln\Big(\textstyle\sum_{j \in J_t} \exp\big(\phi_{ijt} - \tau_{jt} + \beta \mathbb{E}[EV_i(j, \Omega_{it+1}) \mid \Omega_{it}]\big)\Big).$$

   Assumptions used: i.i.d. type-I extreme-value shocks and Rust's conditional independence (pp. 273–274). Unlike static versions, δ_{it} contains future decisions (p. 275).

5. **State reduction** (Assumption 1, (8)–(9), p. 275): $G_i(\phi_{ikt+1}, \delta_{it+1} \mid \Omega_{it}) = G_i(\phi_{ikt+1}, \delta_{it+1} \mid \phi_{ikt}, \delta_{it})$, estimated as $\delta_{it+1} = \rho_{1i} + \rho_{2i}\delta_{it} + \eta_{it}$ and $\phi_{ijt+1} = \gamma_{1i} + \gamma_{2i}\phi_{ijt} + \gamma_{3i}\delta_{it} + \mu_{it}$, independent normal errors, type-specific parameters. *Assumptions used:* Markov beliefs, called bounded rationality and possibly inconsistent with any supply model (p. 275); policy is not a state (fn. 14).

6. **Integrated Bellman equation** (10), p. 276: $EV_i(\phi_{ikt}, \delta_{it}) = \ln\big(\exp(\delta_{it}) + 1[k \neq 0]\exp(W^k_{it}) + \exp(W^0_{it})\big)$, with, in our shorthand, keep value $W^k_{it} = \phi_{ikt} + \beta \mathbb{E}[EV_i(\phi_{ikt+1}, \delta_{it+1}) \mid \phi_{ikt}, \delta_{it}]$ and exit value $W^0_{it} = \beta \mathbb{E}[EV_i(0, \delta_{it+1}) \mid \delta_{it}]$. *Check:* (10) conditions the exit term on $(\phi_{ikt}, \delta_{it})$, (11)–(12) on δ_{it}; these agree because (8) makes δ_{it+1} depend on δ_{it} only.

7. **Choice probabilities** (11)–(12), p. 276, with $D_{it} = \exp(\delta_{it}) + \exp(W^0_{it}) + 1[k \neq 0]\exp(W^k_{it})$:

$$d^{kj}_{it} = \frac{\exp\big(\phi_{ijt} - \tau_{jt} + \beta \mathbb{E}[EV_i(\phi_{ijt+1}, \delta_{it+1}) \mid \delta_{it}, \phi_{ijt}]\big)}{D_{it}}, \qquad \tilde{d}^k_{it} = \frac{\exp(W^k_{it})}{D_{it}}.$$

   Check (ours): $\sum_{j \in J_t} d^{kj}_{it} = \exp(\delta_{it})/D_{it}$; the market enters timing only through δ_{it}.

8. **Aggregation** (13)–(15), p. 276: $s^D_{jt} = \int \sum_k d^{kj}_{it} s_{ikt-1}\, dP_\upsilon$, $\tilde{s}^H_{kt} = \int \tilde{d}^k_{it} s_{ikt-1}\, dP_\upsilon$, $s_{ijt} = \tilde{s}^H_{ijt} + s^D_{ijt}$. Holdings s_{ikt} evolve endogenously by type. *Assumptions used:* market size is the adult population; initial holdings come from a static model without transaction costs (pp. 276, 279).

9. **GMM** (Assumption 2, (16), p. 277): the innovation $\varsigma_{jt} = \xi_{jt} - \lambda\xi_{jt-1}$ satisfies $\mathbb{E}[Z_{jt}\varsigma_{jt}] = 0$, and the estimator is $\hat{\theta} = \arg\min_\theta G(\theta)' W^{-1} G(\theta)$, with nonlinear parameters $\theta_1 = \{\alpha^p, \lambda, \sigma_{\alpha^{age}}\}$. Scrappage take-up in 1997–1998 enters as heavily weighted micromoments (fn. 17). *Assumptions used:* instruments orthogonal to the innovation, not to ξ; observed next-year prices replace $\mathbb{E}_t[p_{jt+1}]$ (p. 276).

10. **Two share inversions** (17)–(18), p. 279, matching each used product's holding and purchase shares:

$$\hat{\phi}'_{jt} = \hat{\phi}_{jt} + \psi_1\big(\ln \check{s}^H_{jt} - \ln \tilde{s}^H_{jt}(\hat{\phi}_{jt}, \tau_{jt}, \theta)\big),$$
$$(\hat{\phi}_{jt} - \tau_{jt})' = (\hat{\phi}_{jt} - \tau_{jt}) + \psi_2\big(\ln \check{s}^D_{jt} - \ln s^D_{jt}(\hat{\phi}_{jt}, \tau_{jt}, \theta)\big),$$

$\psi_1 = 1 - \beta$, $\psi_2 = (1-\beta)^2$; τ_{jt} follows by difference. Keepers do not pay τ, so holding shares identify $\hat{\phi}_{jt}$ and purchase shares $\hat{\phi}_{jt} - \tau_{jt}$ (pp. 279–280). *Note:* the author calls the argument heuristic (p. 280), with no uniqueness proof (p. 279); new-car τ is not identified (fn. 32).

Results to note. $\alpha_p = -52.37$ (4.64), $\lambda = 0.35$ (0.03) with micromoments; $\alpha_p = -2.04$ (4.52) without transaction costs (Table 4, p. 281). The static price coefficient -9.32 (6.13) is starred as significant at 5% though its t-ratio is about 1.5. Average own-price elasticities: static -3.05, dynamic -4.07 (p. 283); the text says the myopic model "upwardly biases" them, but the static values are smaller in magnitude, so we read it as a bias toward zero. Mean used-car τ, the cost of a hypothetical purchase, falls from about EUR 3200 (1994) to EUR 2000 (2004), without standard errors (p. 284, Fig. 4).

Assumptions: stated and implicit.

- *Stated by the author:* i.i.d. EV1 shocks and conditional independence; one car per consumer; buyer-paid τ; full information; perfectly persistent preferences (pp. 270, 279); Assumptions 1–2; $\beta = 0.9$.
- *Implicit, in our reading:* nothing but τ makes a consumer keep a depreciating car, so unmodeled persistent tastes, liquidity, or selling costs load onto τ; expectations are inconsistent (perfect foresight in ϕ, AR(1) beliefs in EV); no budget constraint; no sensitivity to β reported.

Question about the paper.

- An unobserved persistent keep premium: which way does it bias τ in (17)–(18), and what separates them?

2.1.3 Gavazza and Lanteri (2021): credit shocks and car replacement

Why it is included: it is the only car model in our set with liquid wealth, persistent income risk, and a borrowing limit inside the keep, trade, or scrap decision, so it supplies the budget and feasibility structure Rust-type models lack.

Setup. A continuum of households, annual, one indivisible car each (Sec. 4). *Given state:* bonds b, income w, car status $n \in \{1, \ldots, N+1\}$ ($N+1$ is no car), and a permanent type θ, all observed

by household and researcher; income and depreciation shocks realize before decisions. *Actions:* consumption c, bonds b', car to hold $\tilde{n}$ (keep if $\tilde{n} = n$, else trade or scrap). *Prices:* p_1, p_N fixed; used prices and p_b clear markets. No choice shocks anywhere.

Math walkthrough.

1. **Preferences** (1): $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}, d_{it})$, $\beta \in (0,1)$, with the CRRA–Cobb–Douglas form (Sec. 5, following Berger and Vavra 2015)

$$u(c,d) = \frac{(c^\alpha d^{1-\alpha})^{1-\gamma}}{1-\gamma}, \qquad \alpha = 0.95,\ \gamma = 2.$$

   Assumptions used: expected utility, time separability, Cobb–Douglas aggregation. *Note:* $\gamma = 2$ comes from the Aiyagari range, not car data. The credit-shock mechanism runs through precautionary saving, so results depend on γ.

2. **Car services and depreciation** (2): $d(n, \theta_i) = q_n$ for an owner, θ_i for a non-owner. Qualities $q_1 > \cdots > q_N$, new cars q_1; q_n becomes q_{n+1} with probability π_n; q_N is scrapped. Calibration: $N = 4$, $(\pi_1, \pi_2, \pi_3) = (1/3, 0.1, 0.5)$, $(q_1, q_2, q_3) = (1, 0.3, 0.1)$, $\theta \in \{0, 1\}$ with $\mathbb{P}(\theta = 1) = 0.1$. *Assumptions used:* Markov depreciation on four rungs; no horizontal differentiation.

3. **Transaction cost and income.** A seller pays $\lambda(p_n) = \lambda_0 + \lambda_1 p_n$, $\lambda_0 = 0.03$, $\lambda_1 = 0.15$ (affine; no time or search cost). Income: $\log w_{i,t+1} = \rho \log w_{i,t} + \epsilon_{i,t+1}$, $\epsilon \sim N(-0.5\sigma_w^2, \sigma_\epsilon^2)$, $\rho = 0.9$, $\sigma_\epsilon = 0.2$, on a 3-state Rouwenhorst chain. *Check:* the paper states $\sigma_w = \sigma_\epsilon / \sqrt{1-\rho^2} = 0.63$, but $0.2/\sqrt{0.19} = 0.459$; 0.63 is $0.2/\sqrt{1-\rho}$.

4. **Borrowing limit** (3): $b_{i,t+1} \geq \phi$, $\phi \leq 0$, on a one-period non-contingent bond priced p_b, real rate $r = 1/p_b - 1$; calibration $\phi = -1$ (borrowing up to mean annual income). Taxes satisfy $b_G(1 - p_b) = \tau$ (4). *Assumptions used:* one limit common to all households; no default, no amortizing loan contract.

5. **Bellman and budget** (5)–(6):

$$V(b, w, n; \theta) = \max_{c, b', \tilde{n}} \ u\big(c, d(\tilde{n}, \theta)\big) + \beta \mathbb{E}\big[V(b', w', n'; \theta) \mid \tilde{n}, w\big],$$
$$c + p_{\tilde{n}} + p_b b' + \tau = w + p_n - \lambda(p_n)\, \mathcal{I}(\tilde{n} \neq n) + b.$$

   Check: if $\tilde{n} = n$, $p_{\tilde{n}}$ and p_n cancel, so keeping costs nothing. Written out (our reconstruction), $\mathbb{E}[V \mid \tilde{n}, w] = \sum_{w'} F_w(w, w')[(1 - \pi_{\tilde{n}}) V(b', w', \tilde{n}; \theta) + \pi_{\tilde{n}} V(b', w', \tilde{n}+1; \theta)]$ for $\tilde{n} \leq N-1$. Splitting (5) gives car-specific values $v_{\tilde{n}}$ and $V = \max_{\tilde{n}} v_{\tilde{n}}$. *Assumptions used:* $\tilde{n}$, b' chosen jointly after shocks; n' depends only on $\tilde{n}$, w' only on w.

6. **Feasibility bound** (our reconstruction from (3) and (6)): a trade to $\tilde{n}$ is feasible only if $c > 0$ at $b' = \phi$,

$$b > p_{\tilde{n}} - \big[p_n - \lambda(p_n)\big] + p_b \phi + \tau - w.$$

   The bracket is trade-in equity; tightening ϕ raises the bound by p_b per unit, as does a lower used price p_n for trade-in owners. *Note:* necessary only; with $\gamma = 2$, $u \to -\infty$ as $c \to 0$, so optimal thresholds lie strictly above it.

7. **Threshold policy** (Fig. 6): the car policy $g_n(b, w, n; \theta)$ is deterministic and switches at wealth thresholds in (w, b). Read from the plot, the $q_2 \to q_1$ threshold falls from about 12 at $w = 0.5$ to about 5 at $w = 1.5$; the q_3 threshold sits at or just above $\phi = -1$. Rich households replace q_2 with q_1, mid-wealth ones scrap q_3 for q_2, indebted low-income ones keep q_3 until they deleverage.

Key gap: with no taste shocks, $\mathbb{P}(\text{trade} \mid b, w, n, \theta) \in \{0, 1\}$, identical households act identically, and there is no likelihood. Adding EV1 shocks to each $v_{\tilde{n}}$ would turn the threshold into a logit, as in Rust's (4.11)–(4.14) (Section 2.1.1).

8. **Collateral variant** (11), Sec. 6.5: $b_{i,t+1} \geq \phi_t(\chi_0 + \chi_1 \mathbb{E}_t[p_{n_{i,t+1}} \mid \tilde{n}_{it}])$, with $\chi_1 = 0.85$, $\chi_0 = 0.8$. Borrowing capacity rises with the chosen car's expected resale value; no lender pricing or loan-to-value contract.

9. **Market clearing** (7)–(9): bonds, $\int g_b \, dm = b_G$; new cars, $\int \mathcal{I}(g_n = 1) \, dm = \int dm(b, w, n = 1; \theta) + x$ (sets output x); used $\bar{n} = 2, \ldots, N-1$, $\int \mathcal{I}(g_n = \bar{n}) \, dm \leq \int dm(b, w, \bar{n}; \theta)$, strict only with scrapping, then $p_{\bar{n}} = p_N$. Calibrated, $p_3 = p_N$, so only p_2, p_b are endogenous.

10. **Calibration** (Table 1): nothing is estimated statistically. Literature: $\gamma, \rho, \sigma_\epsilon$. One target each: α to a 5% vehicle expenditure share; β to a 2.5% real rate; ϕ to 10% constrained households; b_G to liquid assets/GDP of 1.78 ($b_G = 1.5$ does not visibly map to it); π_n to a 15-year lifetime and 10% scrappage at 15; q_2, q_3 to 3-year depreciation; $p_1 = 0.45$ to price/income; $p_N = 0.036$ to residual value; $\mathbb{P}(\theta = 1)$ to the no-car share; λ_0, λ_1 to NADA retail minus trade-in spreads. Targets interact (e.g. π_n, q_n).

Results to note. Tightening ϕ from -1 to -0.4 over three years lowers the real rate to 1.9% and cuts new-car sales and scrappage by about 7 and used prices by about 10 percentage points (Fig. 8). Constrained owners postpone scrapping, p_2 falls, and the upgrade cost $p_1 - [p_2 - \lambda(p_2)]$ rises for unconstrained ones, who also delay (Sec. 6.2); without used-market clearing, new sales rise (Fig. 10). The collateral variant nearly doubles the drop (Fig. 16).

Assumptions: stated and implicit.

- *Stated by the authors:* steps 1–5 and 10; fixed p_1 (relaxed in Sec. 6.4); competitive used markets; annual period.
- *Implicit, in our reading:* no choice shocks (never discussed); no default or loan schedule; ϕ stands in for credit supply, not a lender's decision; the no-car margin is set by type, not liquidity ($\theta = 0$ gives $u = -\infty$ without a car, $\theta = 1$ gets $d = q_1$ free).

2.1.4 de Lapparent and Cernicchiaro (2012)

TODO: pending full text. Household-panel keep/replace and mileage model for French one-car households, 2000–2007, with income as a stochastic state (from the abstract). To verify: period length, shock distribution, treatment of β, and how income enters the state.

2.2 C2. Identification and estimation

2.2.1 Magnac and Thesmar (2002): identification of dynamic discrete choice

Why it is included: it states exactly which primitives choice and transition data cannot recover, so it tells a modeler what must be fixed before estimated utilities mean anything, and when β can be estimated.

Setup. A theory paper: no data, simulation, or application. Two periods (a short panel), alternatives $i \in I = \{1, \ldots, K\}$, with K the reference alternative. *Given state:* x (includes time, fn. 2, p. 802), observed by agent and researcher, with discrete finite supports X, X' (DS, p. 804);

shocks $\tilde{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_K)$, observed only by the agent. *Action:* $d = i$. *Data:* $\mathbb{P}(d', x', d \mid x) = \mathbb{P}(d' \mid x')\mathbb{P}(x' \mid x, d)\mathbb{P}(d \mid x)$ (p. 805). The paper calls β the "discount rate"; it is the discount factor.

Math walkthrough.

1. **Bellman equations in deterministic parts** (1)–(2), p. 804. From $v_i(x, \tilde{\varepsilon}) = u_i(x) + \varepsilon_i + \beta\mathbb{E}(\max_j v_j(x', \tilde{\varepsilon}') \mid x, d = i)$, CI removes $\tilde{\varepsilon}$ from the continuation term, so $v_i = v_i^*(x) + \varepsilon_i$ and

$$v_i^*(x) = u_i^*(x) + \beta\mathbb{E}\Big(\max_j (v_j^*(x') + \varepsilon_j') \mid x, d = i\Big).$$

   Assumptions used: AS (pp. 802–803), $u_i = u_i^*(x) + \varepsilon_i$, $\mathbb{E}(\varepsilon_i \mid x) = 0$, G absolutely continuous, utilities otherwise unrestricted; PE (p. 803), beliefs equal true transitions, "quite a strong assumption"; CI (p. 803), $\tilde{\varepsilon}'$ independent of $\tilde{\varepsilon}$ given (x, d), which rules out unobserved heterogeneity (p. 804). Under PE transitions come from data, so the structure is $b = \{u_1^*(X), \ldots, u_K^*(X), v_1^*(X'), \ldots, v_K^*(X'), G, \beta\}$, with choice probabilities (3), p. 805.

2. **Hotz–Miller inversion** (4), p. 806: given G, $v_i^*(\tilde{x}) = v_K^*(\tilde{x}) + q_i(\tilde{p}(\tilde{x}); G)$, $q_K = 0$; in period 2, (5): $(v_1^*, \ldots, v_{K-1}^*)(X') = \varphi(\tilde{p}(X'); v_K^*(X'))$. *Assumption used:* G known (Hotz and Miller, 1993). Choice probabilities give value *differences*, never the level v_K^*.

3. **The function** R, p. 806: $R(z; G) = \mathbb{E}_G \max_i(z_i + \varepsilon_i - \varepsilon_K)$. Adding and subtracting v_K inside the max (Appendix A, pp. 813–814) gives $\mathbb{E}_G(\max_j v_j(x', \tilde{\varepsilon}') \mid x') = v_K^*(x') + R(q(\tilde{p}(x')); G)$. *Assumptions used:* $\mathbb{E}(\varepsilon_K \mid x') = 0$ (AS(i)) and (4) at x'. The continuation value is observed CCPs plus one unknown level.

4. **Lemma 1** (6), p. 806. Substitute step 3 into (2), subtract the Kth equation (Appendix (14)–(15)), and eliminate $v_K^*(x)$. For $i < K$,

$$\begin{aligned} u_i^*(x) = u_K^*(x) &- \beta\big[\mathbb{E}(v_K^*(x') \mid x, d = i) - \mathbb{E}(v_K^*(x') \mid x, d = K)\big] \\ &+ q_i(\tilde{p}(x); G) - \beta\big[\mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = i) - \mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = K)\big]. \end{aligned}$$

   Check: as printed in (6) and (8), the parentheses put the last $\mathbb{E}(\cdot \mid x, d = K)$ term outside β. Appendix (14)–(15) and the estimator $Y_i = q_i(\tilde{p}(x)) - \beta R(q(\tilde{p}(x')))$ (p. 808) imply that β multiplies the whole difference, as written here.

5. **Mapping** (7) **and Proposition 2**, pp. 806–807. (6) is a map

$$(u_1^*, \ldots, u_{K-1}^*)(X) = \phi(\tilde{p}(X), \pi; \beta, G, u_K^*(X), v_K^*(X')),$$

   with $\pi = \mathbb{P}(X' \mid X, I)$; the Kth equation only defines $v_K(X)$ and carries no information. With $C = \{c = (\beta, G, u_K^*(X), v_K^*(X'))\}$: (i) for fixed c, exactly one $(u_1^*, \ldots, u_{K-1}^*)(X)$ is compatible with $\tilde{p}(\tilde{X})$; (ii) for all $c, c' \in C \times C$,

$$\{\phi(.; c), \varphi(.; c), c\} \overset{O}{\Leftrightarrow} \{\phi(.; c'), \varphi(.; c'), c'\}.$$

   Consequence: non-identification is exact; every c, including $\beta = 0$ (Manski's static equivalence, p. 807), fits the data equally well. It is larger if G or β depends on x.

6. **Corollary 3 and the moment condition** (8), p. 807. Rearranging (6), for $i < K$,

$$q_i(\tilde{p}(x); G) - \beta\big[\mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = i) - \mathbb{E}(R(q(\tilde{p}(x')); G) \mid x, d = K)\big] = U_i(x),$$

$$U_i(x) \equiv \big[u_i^*(x) + \beta\mathbb{E}(v_K^*(x') \mid x, d = i)\big] - \big[u_K^*(x) + \beta\mathbb{E}(v_K^*(x') \mid x, d = K)\big].$$

Given β and G the left side is data, so $U_i(X)$ is identified with $v_K^*(X')$ unknown. U_i compares "i now, K tomorrow, then optimal" with "K now and tomorrow, then optimal." *Note:* $U_i = u_i^* - u_K^*$ only if $\beta = 0$; otherwise utility differences need $v_K(X')$. The step 4 parenthesis check applies to (8).

7. **Proposition 4**, p. 809. Restriction: $U_i(x_1) = U_i(x_2)$ for some i and $x_1 \neq x_2$. With G fixed, differencing (8) across x_1, x_2 is linear in β. Our reconstruction (the paper states only linearity), with R short for $R(q(\tilde{p}(x')); G)$:

$$\beta = \frac{q_i(\tilde{p}(x_1); G) - q_i(\tilde{p}(x_2); G)}{\Delta},$$

$$\Delta = \big[\mathbb{E}(R \mid x_1, i) - \mathbb{E}(R \mid x_1, K)\big] - \big[\mathbb{E}(R \mid x_2, i) - \mathbb{E}(R \mid x_2, K)\big].$$

Assumptions used: G fixed; rank condition $\Delta \neq 0$; more restrictions overidentify. Stationarity of U_i over three periods supplies one (p. 809). *Note:* absorbing states and terminal dates are testable but not identifying; the reference alternative must be restricted at every period (Sec. 4.1).

8. **Unobserved types** (Section 5, pp. 810–812). (9) conditions everything on a constant type θ. Restricted case: $K = 2$, alternative 2 absorbing with $u_2^* = v_2^* = 0$, $\theta \in \{0, 1\}$ with share $r(\theta)$ independent of x, and (10): θ affects neither transitions nor G. Lemma 5 (11) writes observed CCPs as mixtures with Bayes weights $p(x, \theta) r(\theta)/s(x)$ in period 2, which is dynamic selection. Proposition 6: given G, β, two whole functions and $r(1)$ are unidentified. Proposition 7: adding $r(1)$, additivity $u_1^*(x, \theta) = u_1^*(x) + \theta$, a terminal condition, and rank conditions just identifies u_1^*; β and $r(1)$ need two more exclusions.

Results to note. The unidentified set is exactly $(\beta, G, u_K^*(X), v_K^*(X'))$. Given β and G, U_i is "the only structural object that can be identified" without value-function levels (p. 813). G is never identified, even with parametric utilities (fn. 7, p. 810). No finite-sample results (p. 813).

Assumptions: stated and implicit.

- *Stated by the authors:* AS; PE; CI (relaxed in Section 5); DS; two periods, extensions asserted, not shown (p. 804); G independent of x (p. 803); Section 5 uses two types, two alternatives, and heterogeneity only in preferences (fn. 8, p. 811).
- *Implicit, in our reading:* the same choice set in every state and no budget constraint; rank conditions hold in the data at hand; in a stationary infinite-horizon model the Kth equation is a fixed point determining v_K^*, so fixing $u_K^*(X)$ may suffice (our reconstruction, not in the paper).

2.2.2 Hotz and Miller (1993): conditional choice probability estimation

Why it is included: it shows that the continuation values in a dynamic logit can be recovered from observed choice probabilities and transitions, so a purchase-timing model can be estimated without solving the dynamic program at every trial parameter.

Setup. A finite-horizon dynamic discrete-choice problem with J actions per period (Sec. 2, pp. 498–500). *Given state:* the history $H_t = (b_0', b_1, \ldots, b_{t-1})'$ of an initial endowment and past outcomes, observed by agent and researcher; shocks $\varepsilon_t = (\varepsilon_{t1}, \ldots, \varepsilon_{tJ})$, observed only by the agent. *Action:* $d_{tj} \in \{0, 1\}$, $\sum_j d_{tj} = 1$. *Objective:* $\mathbb{E}_0 \sum_{t=0}^{T} \sum_j d_{tj}[u_j^*(H_t) + \varepsilon_{tj}]$, with β^t inside u^*. Application:

2,088 couples, 1975 National Fertility Survey, 10,257 couple-years, choosing among no contraception, temporary contraception, and irreversible sterilization (pp. 513–514).

Math walkthrough.

1. **Payoff and valuation** (2.3), (2.6)–(2.8), pp. 499–500: $u_{tj} = u_j^*(H_t) + \varepsilon_{tj}$; $v_j(H_t)$ is expected optimal future utility after choosing j, excluding the current payoff; $p_k(H_t) = \mathbb{P}\{k = \arg\max_j [u_{tj}^* + \varepsilon_{tj} + v_{tj}] \mid H_t\}$. *Assumptions used:* additive shocks, independent of H_t; shock distribution $G(\varepsilon_t \mid H_t)$ known up to parameters, with a density (2.4); finite horizon T.

2. **CCPs as a function of value differences** (3.1)–(3.4), p. 500: $p(H_t) = Q(\nu(H_t), H_t)$, $\nu = (v_{t1} - v_{tJ}, \ldots, v_{t,J-1} - v_{tJ})'$. Only differences enter. *Assumption used:* action J serves as the reference.

3. **Inversion** (Proposition 1, p. 501; proof pp. 522–523): Q is invertible in ν, so $\nu(H_t) = Q^{-1}(p(H_t), H_t)$. The Jacobian is diagonally dominant (A.5)–(A.9). *Assumptions used:* a density with positive cross-partials (full support). For type-I extreme value shocks, $v_{tj} - v_{tJ} = \ln(p_j/p_J) - (u_{tj}^* - u_{tJ}^*)$, the form of (4.11), p. 504.

4. **Selection correction and expected payoff** (3.7)–(3.8), p. 501: $W_j = \mathbb{E}(\varepsilon_{tj} \mid H_t, d_{tj}^o = 1)$ is a function of p_t; for extreme value, $W_j = \gamma - \ln p_j$ (4.12). Then $U(p_t, H_t) = \sum_k p_{tk}[u_{tk}^* + W_k]$.

5. **Valuation in CCPs** (3.12), p. 502:

$$V_j(H_t, p(H_t)) = \sum_{s=t+1}^{T} \sum_{H_s \in \mathcal{A}_s(H_t)} U_s \, F_{tj} \prod_{r=t+1}^{s-1} \Big(\sum_k p_{rk} F_{rk} \Big) = v_j(H_t).$$

   Assumptions used: finite outcomes and horizon; agents' beliefs equal the transitions F_j in the data.

6. **Terminating action** (3.13)–(3.14), p. 502: if J, once chosen, must be chosen forever,

$$V_j(H_t) = \sum_{H_{t+1}} \Big\{ U_{t+1} + V_J(H_{t+1}) + \sum_{k<J} p_{t+1,k} Q_k^{-1}(p_{t+1}, H_{t+1}) \Big\} F_j(H_{t+1} \mid H_t),$$

   so only one-period-ahead CCPs are needed. Sterilization plays this role in the application. *Note:* as printed, (4.8)–(4.9) reverse the sign of $v_1 - v_2$ relative to (4.11); (4.11) is the consistent version.

7. **Estimator** (5.4)–(5.9), pp. 506–507: first stage, cell frequencies (5.7) or kernels (5.12) for CCPs and transitions; second stage, GMM on $\mathbb{E}(z_{nt} \otimes [d_{nt} - P(H_{nt}, \hat{p}, \theta)]) = 0$. *Assumptions used:* no unobserved states (Assumption 4); stationary environment and independence across agents, so no aggregate shocks (Assumptions 5–6, p. 506); all feasible histories sampled (p. 509).

8. **Asymptotics** (Proposition 2, p. 508): $N^{1/2}$-consistent and asymptotically normal; the covariance (5.11) corrects for the first stage. Kernel first stages are asymptotically biased; a combination of estimators with different bandwidths removes the bias ((B.27), p. 527).

Results to note. The reduced form is precise ($\chi^2 = 220$, 33 d.f., Table IV, p. 517), and the first-stage correction to standard errors was negligible (p. 517). All three structural specifications are rejected ($p = 0.0034$ for the least restrictive, p. 522), and β is imprecise: 0.646 (s.e. 0.678) in Model C (Table VI, p. 521). No computing times are reported.

Assumptions: stated and implicit.

- *Stated by the authors:* additive shocks with known distribution; finite horizon and finite outcomes; no unobserved state variables; no aggregate shocks; independence across agents; serially uncorrelated income shocks (fn. 10).
- *Implicit, in our reading:* no persistent unobserved heterogeneity (a type would make the CCPs mix types); cells are populated well enough to estimate $\ln p$ without large error; counterfactuals stay within the observed regime, since regime changes may require solving the model (p. 503).

2.2.3 Arcidiacono and Miller (2011)

TODO: pending full text. CCP estimation with discrete unobserved types, and possibly Markov unobserved states, via EM. To verify: whether it handles both time-invariant types and a Markov latent state with an initial-state distribution, and whether vehicle purchase satisfies its finite-dependence condition. Hotz and Miller (1993) use a *terminating* action (sterilization) to simplify the continuation value; whether a purchase that resets the fleet state plays the analogous role is the question this paper should settle. This determines whether conditional independence can be relaxed with a known estimator.

2.2.4 Aguirregabiria and Suzuki (2014): what normalizations do to counterfactuals

Why it is included: it applies Magnac and Thesmar's non-identification to a binary dynamic choice whose payoff has several cost components, and then states which counterfactuals survive the normalization a modeler must impose and what normalized estimates actually measure.

Setup. A theory paper with one numerical example; no data. *Decision maker:* a single firm (later an oligopoly game). *Action:* $a_t \in \{0,1\}$, inactive or active. *Given state:* incumbency $k_t \equiv a_{t-1}$ (endogenous) and exogenous Markov states $\mathbf{z}_t = (\mathbf{z}_t^v, \mathbf{z}_t^c)$ with transition f_z, both observed by firm and researcher; shocks ε_t observed only by the firm. *Data:* many markets, short panels; β, f_z and variable profit vp are treated as known (Sec. 3.1, p. 276). Fixed profit has three unknown functions of $\mathbf{z}^c$: fixed cost fc, entry cost ec, scrap value sv.

Math walkthrough.

1. **Payoffs and choice rule** (3), (5)–(10), pp. 272–274:

$$\Pi_t = \begin{cases} k_t\, sv(\mathbf{z}_t^c) & a_t = 0, \\ vp(\mathbf{z}_t^v) - fc(\mathbf{z}_t^c) - (1-k_t)\, ec(\mathbf{z}_t^c) & a_t = 1, \end{cases}$$

   plus $\varepsilon_t(0) = k_t \varepsilon_t^{sv}$, $\varepsilon_t(1) = -\varepsilon_t^{fc} - (1-k_t)\varepsilon_t^{ec}$. With conditional value $v(a,k,\mathbf{z})$ and $\tilde{v}(k,\mathbf{z}) = v(1,k,\mathbf{z}) - v(0,k,\mathbf{z})$, the firm is active iff $\tilde{\varepsilon}_t \le \tilde{v}(k_t,\mathbf{z}_t)$, so $P(k,\mathbf{z}) = F_{\tilde{\varepsilon}|k}(\tilde{v}(k,\mathbf{z}))$. *Assumptions used:* additive shocks, i.i.d. over time, independent of $(\mathbf{z}_t, k_t)$, zero mean (p. 272); no shock in variable profit (p. 273); f_z independent of a_t; the inactive non-incumbent earns zero, $\pi(0,0,\mathbf{z}) = 0$ ((3), p. 272).

2. **CCP inversion** (16), p. 278: $\tilde{v}(k,\mathbf{z}) = F^{-1}_{\tilde{\varepsilon}|k}(P(k,\mathbf{z}))$. *Assumptions used:* $F_{\tilde{\varepsilon}|k}$ strictly monotone and identified by Proposition 1 (p. 277), which needs a *finitely lived* firm and vp_t as a special regressor with continuous support; no infinite-horizon proof is known to the authors (fn. 13). Without price data everything is identified only up to an unknown scale (pp. 277–278).

3. **Continuation difference in observables** (17)–(19), pp. 278–279. With $\bar{V}(k,\mathbf{z}) = \int V\, dG$, (18) gives $\bar{V}(k,\mathbf{z}) = v(0,k,\mathbf{z}) + S(\tilde{v}(k,\mathbf{z}), F_{\tilde{\varepsilon}|k})$, $S(\tilde{v},F) \equiv \int_{-\infty}^{\tilde{v}} [\tilde{v}-\tilde{\varepsilon}]\, dF(\tilde{\varepsilon})$, a known function. Since both $(a,k) = (0,1)$ and $(0,0)$ lead to $k=0$ next period, $v(0,1,\mathbf{z}') - v(0,0,\mathbf{z}') = sv(\mathbf{z}^{c\prime})$ (p. 279). *Assumptions used:* $\mathbb{E}\varepsilon(0) = 0$; $\mathbf{z}$ exogenous. Our reconstruction of the flow difference: $\pi(1,k,\mathbf{z}) - \pi(0,k,\mathbf{z}) = vp - fc - ec + k(ec - sv)$.

4. **The identified function** Q (20), p. 279. Subtracting the known S terms from $\tilde{v}$ defines $Q(k,\mathbf{z})$, and

$$Q(k,\mathbf{z}) = vp(\mathbf{z}^v) - [fc(\mathbf{z}^c) + ec(\mathbf{z}^c)] + k\,[ec(\mathbf{z}^c) - sv(\mathbf{z}^c)] + \beta \textstyle\sum_{\mathbf{z}^{c\prime}} f_z(\mathbf{z}^{c\prime} \mid \mathbf{z})\, sv(\mathbf{z}^{c\prime}).$$

   Q is "active today, exit next period" minus "exit today" (p. 279). In our reading it is Magnac and Thesmar's $U_i(x)$ (their (8); step 6 of Section 2.2.1) with "inactive" as reference.

5. **Proposition 2: non-identification** (21), pp. 279–280; proof p. 301. Two equations ($k=0,1$) per $\mathbf{z}$ in three functions identify only

$$ec - sv = Q(1,\mathbf{z}) - Q(0,\mathbf{z}), \qquad fc + ec - \beta \textstyle\sum f_z\, sv(\mathbf{z}^{c\prime}) = vp(\mathbf{z}^v) - Q(0,\mathbf{z}).$$

   For any $\lambda(\mathbf{z}^c)$, $(ec{+}\lambda,\ sv{+}\lambda,\ fc{-}\lambda{+}\beta\mathbb{E}[\lambda(\mathbf{z}^{c\prime}) \mid \mathbf{z}])$ fits the same data. *Check* (our reconstruction): this is the payoff shift $k\lambda(\mathbf{z}) - \beta a\mathbb{E}[\lambda(\mathbf{z}') \mid \mathbf{z}]$ with $V^* = V + k\lambda$, the case $\Phi = k\lambda$ of adding $\Phi(\text{state}) - \beta\mathbb{E}[\Phi(\text{next state}) \mid \text{state, action}]$ to every payoff, which leaves all CCPs unchanged for fixed β and transitions.

6. **What normalized estimates measure** (22) and Table 1, pp. 280–281. Under $\widehat{sv} = 0$: $\widehat{ec} = ec - sv$ (ex-ante sunk entry cost) and $\widehat{fc} = fc + sv - \beta\mathbb{E}[sv(\mathbf{z}^c_{t+1}) \mid \mathbf{z}_t]$. Under $\widehat{fc} = 0$: $\widehat{ec}$ and $\widehat{sv}$ each add $\sum_{r\geq 0} \beta^r \mathbb{E}[fc(\mathbf{z}^c_{t+r}) \mid \mathbf{z}_t]$, the present value of operating forever (p. 281). Under $\widehat{ec} = 0$: $\widehat{sv} = sv - ec$, $\widehat{fc} = fc + ec - \beta\mathbb{E}[ec(\mathbf{z}^c_{t+1}) \mid \mathbf{z}_t]$. *Check* (our reconstruction of row 2): (22) gives the recursion $\widehat{sv} - sv = fc + \beta\mathbb{E}[(\widehat{sv} - sv)']$, a contraction for $\beta < 1$, bounded fc.

7. **Counterfactuals: Lemma 1 and Propositions 3–5**, pp. 284–286; proofs pp. 300–302. Lemma 1: for given (β, f_z) the map from the whole CCP vector to the whole Q vector, (23), is one-to-one, so P^* is identified iff Q^* is; in our reading, the Jacobian's nonsingularity is asserted without the full argument. Split $\mathbf{z} = (\mathbf{z}^{nosv}, \mathbf{z}^{sv})$, $\mathbf{z}^{sv}$ the states entering sv. Differencing (20) gives (24), printed as $Q^* = Q^0 + \text{RHS}$; with $\Delta_Q \equiv Q^* - Q^0$ (our notation),

$$\begin{aligned}\Delta_Q(k,\mathbf{z}) = \Delta_{vp} - [\Delta_{fc} + \Delta_{ec}] + k[\Delta_{ec} - \Delta_{sv}] + \beta^0 \textstyle\sum f^0_{z,sv}\, \Delta_{sv}(\mathbf{z}^{sv\prime}) \\ + \beta^0 \textstyle\sum \Delta_{f_{z,sv}}\, [sv^0 + \Delta_{sv}](\mathbf{z}^{sv\prime}) + \Delta_\beta \textstyle\sum [f^0_{z,sv} + \Delta_{f_{z,sv}}]\, [sv^0 + \Delta_{sv}](\mathbf{z}^{sv\prime}).\end{aligned}$$

   Prop. 3 (p. 285): with $\Delta_\beta = 0$, $\Delta_{f_{z,sv}} = 0$ and known $\{\Delta_{vp}, \Delta_{fc}, \Delta_{ec}, \Delta_{sv}, \Delta_{f_{z,nosv}}\}$, the second line vanishes and P^* is identified. Prop. 4 (p. 285): if $\Delta_\beta \neq 0$ or $\Delta_{f_{z,sv}} \neq 0$, Δ_Q depends on the unidentified level sv^0, (30), even with factual and counterfactual (β, f_z) known. Prop. 5 (p. 286): using $\widehat{sv^0}$ for sv^0, the bias (25) is the second line with $[\widehat{sv^0} - sv^0]$ in place of $[sv^0 + \Delta_{sv}]$, zero iff $\Delta_\beta = \Delta_{f_{z,sv}} = 0$. *Check:* (25) sums over $\mathbf{z}^{sv\prime}$, valid when $\widehat{sv^0} - sv^0$ depends only on $\mathbf{z}^{sv}$, as under $\widehat{sv} = 0$; under $\widehat{fc} = 0$ the gap depends on every state predicting future fc, so the "innocuous" class is narrower than stated, although the text allows normalizations "not necessarily" $\widehat{sv^0} = 0$.

8. **Identifying conditions** (Sec. 5, pp. 293–296). Bounds from $ec, fc, sv \geq 0$ and $sv \geq \beta\mathbb{E}sv'$ (Sec. 5.1, p. 293); *Check:* printed $sv_{LOW} = -ec_{LOW} \leq 0$ is weaker than the imposed $sv \geq 0$. Prop. 6 (p. 294): if $\mathbf{z}^c_2$ is excluded from fc and from one of ec, sv, and shifts the transition of $\mathbf{z}^c_1$ with rank $|\mathcal{Z}^c_1| - 1$, the three functions are identified up to a constant; the exclusion is untestable

(p. 295). Prop. 7 (p. 295): transition counterfactuals then identified; β-counterfactuals still not. Prop. 8 (p. 296): with acquisition-only entry/exit at a Nash-bargained price (26), classical price error and known bargaining power α, (27) identifies the components; fn. 19 expects selection to make the data reject this.

Results to note. Calibrated land-price example (Sec. 4.3, pp. 287–293): giving high-price markets the low-price process truly shifts entry and stay schedules *down*; under $\widehat{sv} = 0$ they shift *up* (Fig. 4, p. 292), and a potential entrant's unconditional probability of being active is overstated by 4.7 percentage points (15.1 vs. 10.4), p. 292. One illustration chosen to show bias, not evidence of typical magnitude. Table 2 (p. 286): the only transition-change applications are Collard-Wexler (2013) and Das et al. (2007), both under $sv = 0$. *Check*, arithmetic slips in print: Example 2's $\widehat{ec} = 0$ case omits the intercept $(1-\beta)ec_0$ (p. 282); "$5.6 + 0.4z^c$" should be $5.6 + 0.04z^c$ (p. 289); the stationary s.d. of the stated AR(1) is ≈ 1.15, not 1.3 (p. 288).

Assumptions: stated and implicit.

- *Stated by the authors:* additive i.i.d. shocks independent of $(\mathbf{z}, k)$, no variable-profit shock (pp. 272–273); β, f_z known and *vp* identified from price/quantity data (p. 276); shock distribution identified only for a finitely lived firm (Prop. 1, fn. 13); $\pi(0, 0, \mathbf{z}) = 0$; large number of markets, short T (p. 276); serial correlation has no "substantive influence" (pp. 272–273), asserted, not shown; the game adds only over-identifying restrictions (pp. 282–283).
- *Implicit, in our reading:* beliefs equal the true f_z; a counterfactual change must be known in utility units and stated in levels, not as a percentage of a normalized component; persistent heterogeneity would need the argument type by type with the type mixture identified elsewhere (Kasahara and Shimotsu, 2009); Prop. 1's finite-horizon restriction leaves the infinite-horizon results without a proved money anchor.

Question for discussion. If replacing a vehicle is read as a simultaneous exit (resale, the analog of *sv*) and entry (outlay, the analog of *ec*), does a purchase-outlay counterfactual that also changes the post-purchase liquidity transition fall in Proposition 4's non-identified class?

2.2.5 Arcidiacono and Miller (2020): identification and counterfactuals off short panels

Why it is included: it characterizes every payoff vector that fits the same choice data, shows what a *known* (not normalized) payoff buys, and states which counterfactuals are identified without one, which is the question any counterfactual from a purchase-timing model must answer.

Setup. A theory paper with one two-period example; no data. *Decision maker:* a single agent with horizon $T \leq \infty$; the data end at S, and a "short panel" means $S < T$ in a nonstationary model (p. 473). *Action:* $j \in \{1, \ldots, J\}$, indicators d_{jt}, mutually exclusive (Assumption 1). *Given state:* discrete $x_t \in \{1, \ldots, X\}$, observed by agent and researcher, with transitions $f_{jt}(x_{t+1} \mid x_t)$ (F_j when stationary); shocks ϵ_t with density g, seen only by the agent. *Data:* CCPs $p_{jt}(x)$ for $t \leq S$. T, β, f, g are known (p. 476).

Math walkthrough.

1. **Problem and CCP correction** (1)–(4), pp. 475–476. Conditional value (3): $v_{jt}(x_t) = u_{jt}(x_t) + \beta \sum_{x_{t+1}} V_{t+1}(x_{t+1}) f_{jt}(x_{t+1} \mid x_t)$; $\psi_{jt}(x) \equiv V_t(x) - v_{jt}(x)$, identified from CCPs given g; choose j iff $\epsilon_{kt} - \epsilon_{jt} \leq \psi_{kt} - \psi_{jt}$ for all k (4). The paper states $\psi_{jt} = -\ln p_{jt}$ for standard EV1 (pp. 476, 480). *Assumptions used:* additive, conditionally independent shocks with continuous support; bounded payoffs (Assumptions 2–3, p. 475). *Check:* $-\ln p$ holds for mean-zero EV1; with mean γ, $\psi = \gamma - \ln p$, and the constant cancels wherever ψ enters through differences.

2. **CCP representation** (5)–(6), p. 476, and (10), p. 477. Replacing V_{t+1} by $v_{l,t+1} + \psi_{l,t+1}$ for a normalizing action $l(x,\tau)$ and repeating,

$$v_{jt}(x_t) = u_{jt}(x_t) + \sum_{\tau=t+1}^{T} \sum_{x} \beta^{\tau-t} \big[u_{l(x,\tau),\tau}(x) + \psi_{l(x,\tau),\tau}(x)\big] \kappa^*_{\tau-1}(x \mid x_t, j),$$

with κ^* the state distribution after j at t and l thereafter; (10) is the case $l = 1$. *Assumptions used:* the definition of ψ and known f. In our reconstruction, nothing uses optimality of l, so any action may serve.

3. **Theorem 1 and Corollary 2: the equivalence class** (7)–(9), pp. 476–477; proofs pp. 481–483. For given (T, β, f, g), choosing in each (x,t) a normalizing action and an arbitrary bounded payoff $c_t(x)$ for it generates exactly the observationally equivalent payoffs. Stationary, (9):

$$u^*_j = u_j + [\mathcal{I} - \beta F_j][\mathcal{I} - \beta F_l]^{-1}(c - u_l).$$

With $h \equiv [\mathcal{I} - \beta F_l]^{-1}(c - u_l)$ this is $u^*_j = u_j + h - \beta F_j h$ (our rewriting), so $V^* = V + h$ and CCPs are unchanged. One free payoff per state (and period). The usual reading of normalized payoffs as "net of" action 1's is "grossly misleading, if not false" outside a static model (p. 477): they also absorb differences in action 1's future payoffs along the two paths.

4. **Theorem 3: inversion off long panels** (11)–(12), pp. 477–478; proof p. 483. If $u_{1t}(x)$ is *known* for all (x,t), differencing (10) for j and 1 gives

$$\begin{aligned} u_{jt}(x) = {} & u_{1t}(x) + \psi_{1t}(x) - \psi_{jt}(x) \\ & + \sum_{\tau=t+1}^{T} \sum_{x_\tau} \beta^{\tau-t} \big[u_{1\tau}(x_\tau) + \psi_{1\tau}(x_\tau)\big] \big[\kappa_{\tau-1}(x_\tau \mid x, 1) - \kappa_{\tau-1}(x_\tau \mid x, j)\big]. \end{aligned}$$

Check: the page prints $\psi_{1t}(x_\tau)$ inside the sum; the proof and (16) use $\psi_{1\tau}(x_\tau)$. Stationary (12) is printed as $u_j = \Psi_j - \Psi_1 - u_1 + \beta(F_1 - F_j)[\mathcal{I} - \beta F_1]^{-1}(\Psi_1 + u_1)$; the first three terms have the wrong sign. The proof (p. 483) derives $u_j = u_1 + \Psi_1 - \Psi_j + \beta(F_1 - F_j)[\mathcal{I} - \beta F_1]^{-1}(u_1 + \Psi_1)$, which our numerical check reproduces and printed (12) does not. The systems are exactly identified; plug-in estimators are claimed asymptotically efficient (argued, not proved), and "false restrictions, such as adopting convenient functional forms for the payoffs, typically create misspecifications" (p. 478).

5. **Single-action finite dependence** (13), p. 478. If repeating action 1 makes the state distribution independent of the initial action after ρ periods (terminal actions, renewal, multiperiod renewal), the sum in step 4 stops at $t + \rho$, and "if (13) holds and $u_{1t}(x_t)$ is known for all $t \le S$, then $u_{jt}(x_t)$ is identified for all $t \le S - \rho$." The payoff must be "actually known, not just normalized." *Check:* read literally, (13) compares states after $\rho - 1$ repetitions and would exclude the bus-engine renewal the text includes; we read ρ as the number of repetitions.

6. **Corollary 4: short panels** (14), p. 479. Using the known u_1 only up to S,

$$u_{jt}(x) = \ldots + \beta^{S-t} \sum_{x_{S+1}} V_{S+1}(x_{S+1}) \big[\kappa(x_{S+1} \mid x, 1) - \kappa(x_{S+1} \mid x, j)\big],$$

where "…" is step 4's expression with the sum truncated at S. Flows are linear in the unknown V_{S+1}, which "depends on CCPs that occur after the sample ends"; in general X further restrictions are needed. *Check:* the κ difference sums to zero, so a constant in V_{S+1} drops and $X - 1$ suffice; the exponent β^{S-t} is as printed.

7. **Theorem 5: temporary payoff innovations**, pp. 479–480; proof pp. 483–484. If $\Delta_{jt}(x) = \tilde{u}_{jt}(x) - u_{jt}(x) = 0$ for $t > S$, then $\tilde{p}_{S+1} = p_{S+1}$, $\tilde{p}_{jS}$ follows from $\psi_{kS} - \psi_{jS}$ shifted by $\Delta_{jS} - \Delta_{kS}$, and backward for $t < S$,

$$\tilde{\psi}_{kt} - \tilde{\psi}_{jt} = \psi_{kt} - \psi_{jt} + \Delta_{jt} - \Delta_{kt} + \sum_{\tau=t+1}^{S} \sum_{x_\tau} \beta^{\tau-t} [\Delta_{1\tau} + \tilde{\psi}_{1\tau} - \psi_{1\tau}](x_\tau) [\kappa_{\tau-1}(x_\tau \mid x, j) - \kappa_{\tau-1}(x_\tau \mid x, k)].$$

   Only Δ_1 enters, never u_1, so the counterfactual needs no known payoff. The same holds for temporary changes in g (fn. 6). Permanent policies off short panels are not identified (p. 479).

8. **Transition innovations** (15), Sec. 4.2, p. 480. With counterfactual $\tilde{\kappa}$, the $u_{1\tau}$ terms no longer cancel because the paths' state weights change: the counterfactual "cannot be derived without knowing the true systematic payoff for one of the choices, regardless of the sample length." Off short panels it also needs single-action finite dependence; otherwise $\tilde{p}_{jt}$ is "not identified for any t." Argued in text, without a numbered theorem. Our stationary reconstruction: a data-consistent $u_j^* = u_j + h - \beta F_j h$ evaluated under $\tilde{F}_j$ carries the extra payoff $\beta(\tilde{F}_j - F_j)h$, zero iff $\tilde{F} = F$. The paper attributes the stationary payoff-versus-transition result to Aguirregabiria (2005, 2010) and Norets and Tang (2014) (pp. 474, 479).

Results to note. Smoking example (Sec. 5, pp. 480–481), $u_{1t}(1) = 0$, $u_{1t}(2) = c$: normalizing $u_{1t}^* \equiv 0$ biases the smoking payoff by $\beta\pi c$ (17); a subsidy Δ to the sick gives $\tilde{p}_{11}(1) = p_{11}(1)/\{p_{11}(1) + [1 - p_{11}(1)]\exp(\beta\Delta\pi)\}$ from period-1 CCPs alone; a change $\pi \to \tilde{\pi}$ under the normalization is off by the factor $\exp[\beta(\pi - \tilde{\pi})c]$. *Check:* the subsidy is paid at $t = 2 > S = 1$, so it lies outside Theorem 5's literal hypothesis (the result holds because the payment is common to both actions); the printed base $[p_{12}(1)/p_{12}(2)]$ should be inverted, which our numerical check confirms. Conclusion (p. 481): estimating utilities "purely as a vehicle for making counterfactual predictions is not compelling" without outside knowledge of the utility functions. No finite-sample results.

Assumptions: stated and implicit.

- *Stated by the authors:* Assumptions 1–3 (p. 475): mutually exclusive actions, finite discrete states, additive conditionally independent shocks with continuous support, bounded payoffs; T, β, f, g known, g "extreme value for tractability" (p. 476, fns. 3–4); short-panel counterfactuals restricted to temporary innovations (p. 479); transition counterfactuals need a payoff known from outside the data (Sec. 4.2); persistent heterogeneity left to other work (p. 474), e.g. Kasahara and Shimotsu (2009).

- *Implicit, in our reading:* CCPs are known in every sampled (x, t) cell, so precision with sparse cells is not addressed; beliefs equal realized transitions; how beliefs about aggregate states enter is not specified (p. 479 only excludes them); in a stationary model short histories matter through state coverage and initial conditions (Section 2.5.1), not through Corollary 4.

Question for discussion. In a stationary keep/replace model where replacement lowers rather than resets a liquidity state, replace is not a renewal action; is a purchase-outlay counterfactual identified without a known payoff only when it shifts the replace payoff by a known amount in utils and leaves the liquidity transition unchanged?

2.2.6 Kasahara and Shimotsu (2009): identifying finite mixtures of dynamic discrete choices

We work from the authors' working paper (Queen's Economics Department Working Paper No. 1092, which also covers estimation), not the published article, so equation, proposition, remark and page numbers are working-paper locations, and numbering and conditions may differ from the 2009 article. *Why it is included:* it gives sufficient conditions under which permanent unobserved types, their initial distributions, and their choice probabilities are identified nonparametrically from a short panel, the result usually invoked to justify a finite-type household model.

Setup. Theoretical econometrics, with a series-logit estimator (Sec. 4) and a Monte Carlo on a Rust bus model (Sec. 5). M types, known (p. 25), with probabilities π^m; the type is unobserved by the researcher. *Observed:* state $x_t \in X$ and action $a_t \in A = \{0, 1\}$, $t = 1, \ldots, T$, for a random sample of units, $N \to \infty$, T fixed (p. 6). No utilities, β or shocks are used; a forward-looking structural model is a special case (Example 2, (6)–(7), p. 7). Identification is up to relabelling of types (p. 8).

Math walkthrough.

1. **Mixture** (3), p. 6:

$$P(\{a_t, x_t\}_{t=1}^T) = \sum_{m=1}^M \pi^m p^{*m}(x_1, a_1) \prod_{t=2}^T f^m(x_t \mid x_{t-1}, a_{t-1})\, P^m(a_t \mid x_t, a_{t-1}).$$

   Assumptions used: finite M; the type-specific initial distribution $p^{*m}(x_1, a_1)$ is unrestricted and need not be stationary (Remark 1, p. 6), which is the setting of Heckman (1981) (Example 1, p. 7).

2. **Baseline model and division by transitions** (8)–(9), p. 8. Assumption 1: (a) $P^m(a_t \mid x_t, a_{t-1}) = P^m(a_t \mid x_t)$; (b) $f^m(x' \mid x, a) > 0$ for all (x', x, a); (c) $f^m = f$; CCPs are time-invariant. Then f is identified from observed (x', x, a) and

$$\tilde{P}(\{a_t, x_t\}_{t=1}^T) \equiv \frac{P(\{a_t, x_t\}_{t=1}^T)}{\prod_{t=2}^T f(x_t \mid x_{t-1}, a_{t-1})} = \sum_{m=1}^M \pi^m p^{*m}(x_1, a_1) \prod_{t=2}^T P^m(a_t \mid x_t),$$

   a mixture of products: given type, periods are independent. *Assumptions used:* (c) lets f leave the sum; (b) makes the division defined for every sequence.

3. **Submodels** (10)–(11), p. 9: dropping periods gives the same form over any subset $\mathcal{I}$ of periods, with p^{*m} if $1 \in \mathcal{I}$ and without it otherwise. *Our reconstruction:* a period $t \geq 2$ is removed by summing $\tilde{P}$ over a_t at a fixed, arbitrary x_t, which requires (b).

4. **Moments and matrices** (12)–(16), pp. 9–11. Fix $a_t = 1$: $\lambda_\xi^m = P^m(a = 1 \mid x = \xi)$, $\lambda_k^{*m} = p^{*m}((a_1, x_1) = (1, k))$; moments such as $F_{x_2,x_3} = \sum_m \pi^m \lambda_{x_2}^m \lambda_{x_3}^m$ and $F^*_{x_1,x_2,x_3} = \sum_m \pi^m \lambda_{x_1}^{*m} \lambda_{x_2}^m \lambda_{x_3}^m$ fill two $M \times M$ matrices P, P_k with $x_1 = k$, $x_2, x_3 \in \{\xi_1, \ldots, \xi_{M-1}\}$, and a first row and column from the submodels that drop a period. Proof (p. 31): $P = L'VL$ and $P_k = L'VD_kL$, with L the $M \times M$ matrix with rows $(1, \lambda_{\xi_1}^m, \ldots, \lambda_{\xi_{M-1}}^m)$, $V = \mathrm{diag}(\pi^m)$, $D_k = \mathrm{diag}(\lambda_k^{*m})$.

5. **Proposition 1** (pp. 9–10; proof pp. 31–32). If $T \geq 3$, L is nonsingular for some $\xi_1, \ldots, \xi_{M-1} \in X$, and some $k \in X$ has λ_k^{*m} distinct across types, then $\{\pi^m, \lambda_\xi^{*m}, \lambda_\xi^m\}$ is uniquely determined. Construction: $P^{-1}P_k = L^{-1}D_kL$, so the eigenvalues are λ_k^{*m} and the eigenvectors are the columns of L^{-1} up to scale; the column of ones in L fixes the scale; the first column of $(L')^{-1}P = VL$ is $(\pi^1, \ldots, \pi^M)'$; one good set of $M-1$ points gives λ_ζ^m at every other ζ via (29), p. 32 (Remark 2.3).

Assumptions used: L nonsingular means "different types must respond to [the state's] changes differently" (Remark 2.1, p. 10); for $M = 2$, $\det L = \lambda^2_{\xi_1} - \lambda^1_{\xi_1}$. Distinct eigenvalues make the eigenvectors unique up to scale.

6. **Corollary 1** (p. 11), the data version: P full rank and $P^{-1}P_k$ with distinct eigenvalues, checkable from sample moments.

7. **Panel length and number of types.** $T \geq 3$ is needed without further restrictions: at $T = 2$ parameters outnumber restrictions (p. 25). $T \geq 4$ if initial distributions do not differ by type (Remark 2.2, p. 10). $M \leq |X| + 1$ under Proposition 1 (Remark 2.4). Proposition 2 (p. 12): odd $T \geq 3$, $u = (T-1)/2$, $\sum_{l=0}^{u} \binom{|X|+l-1}{l} \geq M$ via Hadamard products.

8. **Extensions** (Sec. 3). Time-varying CCPs: $T \geq 3$ with L_t nonsingular every period (Proposition 3, p. 14). Type-specific transitions: with $Q^m(s_t \mid s_{t-1}) = f^m(x_t \mid x_{t-1}, a_{t-1})P^m(a_t \mid x_t)$, fixing $s_t = \bar{s}$ in odd periods restores a product, requiring $T \geq 6$ and $G_{\bar{s}}$ nonsingular, $\gamma^m_{\bar{s}}(s) = Q^m(\bar{s} \mid s)Q^m(s \mid \bar{s})$ (Proposition 5, p. 16). Lagged choice in the CCP: $T \geq 6$ (Sec. 3.3, p. 19). An extra state with type-common full-support transitions: $T \geq 4$ (Proposition 7, p. 18). Limited transitions: only prefixes are available; Proposition 8 ($T = 4$, pp. 22–23) uses all-$a_t = 0$ paths through sets with $B^h \subseteq \Gamma(0, \{h\})$, $C^h \subseteq \Gamma(0, B^h) \cap \Gamma(0, \{h\})$, $\{k\} \subseteq \Gamma(0, C^h) \cap \Gamma(0, B^h) \cap \Gamma(0, \{h\})$, where $\Gamma(a, W) = \bigcap_{x \in W} \{x' : f(x' \mid x, a) > 0\}$ (Assumption 4(b), p. 22). Local identification by full column rank of the Jacobian (Proposition 10, p. 25). *Check:* Proposition 3 defines ξ^t_j for $t \leq T - 1$ but requires L_t for $t \leq T$ (p. 14); Proposition 9's "$T \geq 4$ is odd" means $T \geq 5$ (p. 23).

9. **Estimator** (Sec. 4, pp. 25–28): series logit for P^m; Lemma 1 gives CCP rate $N^{-1/5}$ when $K = N^{1/5}$, $s/r = 4$. *Assumptions used:* continuous compact x; CCPs bounded away from 0 and 1; a *stationary* initial distribution (Assumption 7, pp. 26–27).

Results to note. The theorems are population results with no conditioning analysis. Table 2 (p. 43; two types, $\pi = (0.5, 0.5)$, 100 replications): RMSE of $\hat{\pi}$ at $(N, T) = (500, 3)$ is 0.112 (series) and 0.130 (MLE) against 0.025 with types observed; at $(500, 10)$, 0.041 and 0.043 against 0.024. In our reading, the Monte Carlo design ($T = 3$, limited and, in Table 5, type-specific transitions) lies outside the sufficient conditions of Propositions 5 and 8, so its success rests on the parametric structure.

Assumptions: stated and implicit.

- *Stated by the authors:* finite, known M; Assumption 1(a)–(c); time-invariant CCPs (relaxed in Proposition 3); binary A (multinomial "straightforward," not shown, p. 9); random sampling of complete sequences (p. 6); Assumptions 7–8 for the estimator.

- *Implicit, in our reading:* a balanced panel without type-dependent attrition; a nonsingular L with a determinant near zero satisfies every proposition, so weak identification is invisible; full support excludes deterministic states such as age or tenure; the estimator's stationary initial distribution contradicts Remark 1; identifying P^m leaves the Magnac and Thesmar (2002) normalizations to be imposed type by type (fn. 5, p. 5).

Question for discussion. If vehicle age moves deterministically (one month older on keep, zero on replace), Assumption 1(b) fails, the next state reveals the action so the middle-period submodels of step 3 do not exist, and in our reading for the all-keep paths that Proposition 8 uses its set C^h is empty below the age cap: does any result here apply, or would type identification rest on parametric structure?

2.2.7 Higgins and Jochmans (2023): mixtures of Markov chains from four observations

Why it is included: it identifies permanent types, their initial distributions, and type-specific transitions from four consecutive observations without the full-support transitions that Kasahara and Shimotsu (2009) require (Section 2.2.6), so deterministic state variables are not excluded in form.

Setup. Theoretical econometrics, with an EM estimator and a small Monte Carlo (Appendix B). A latent type $Z \in \{1, \dots, q\}$, q known, with $\mu_z > 0$ (Sec. 2, p. 2). *Observed:* $X_t \in \{1, \dots, r\}$, the choice and state enumerated into one variable (pp. 2, 8). Given $Z = z$, X_t is a first-order, time-homogeneous Markov chain with initial distribution $s_z(x) = \mathbb{P}(X_1 = x \mid Z = z)$, not assumed stationary, and kernel $k_z(x, x') = \mathbb{P}(X_t = x' \mid X_{t-1} = x, Z = z)$. The kernel absorbs choice probabilities, state transitions and lagged-choice dependence, all type-specific and unrestricted. No behavioral model. Identification is up to one common relabelling of types (p. 2).

Math walkthrough.

1. **Observables and factorization** (3.1), pp. 3–4. With $(\boldsymbol{P}_x)_{i,j} = p_{j,x,i} = \mathbb{P}(X_1 = j, X_2 = x, X_3 = i)$ and $(\boldsymbol{P}_{x,x'})_{i,j} = p_{j,x,x',i}$,

$$\boldsymbol{P}_x = \boldsymbol{K}_x \boldsymbol{L}_x^\top, \qquad \boldsymbol{P}_{x,x'} = \boldsymbol{K}_{x'} \boldsymbol{D}_{x,x'} \boldsymbol{L}_x^\top,$$

   $(\boldsymbol{K}_x)_{x',z} = k_z(x, x')$, $(\boldsymbol{L}_x)_{x',z} = \mu_z s_z(x') k_z(x', x)$, $\boldsymbol{D}_{x,x'} = \operatorname{diag}(k_1(x, x'), \dots, k_q(x, x'))$. *Our reconstruction:* $p_{j,x,i} = \sum_z \mu_z s_z(j) k_z(j, x) k_z(x, i)$ by total probability over Z; given $X_2 = x$ and Z, past and future are independent, and a fourth period inserts one diagonal factor $k_z(x, x')$. *Assumptions used:* first-order Markov given type; time homogeneity, so the same k_z appears in every transition.

2. **Assumption 1** (p. 3): "For each x the $r \times r$ matrix $\boldsymbol{P}_x$ has rank q." Equivalent to $\boldsymbol{K}_x$ and $\boldsymbol{L}_x$ both of column rank q; necessary $r \geq q$; the authors call it "directly testable from the data."

3. **Whitening** (3.2)–(3.3), p. 4. From the thin SVD $\boldsymbol{P}_x = \boldsymbol{U}_x \boldsymbol{E}_x \boldsymbol{V}_x^\top$, set $\boldsymbol{A}_x = \boldsymbol{E}_x^{-1/2} \boldsymbol{U}_x^\top$, $\boldsymbol{B}_x = \boldsymbol{E}_x^{-1/2} \boldsymbol{V}_x^\top$, so $\boldsymbol{A}_x \boldsymbol{P}_x \boldsymbol{B}_x^\top = \boldsymbol{I}_q$. With $\boldsymbol{Q}_x = \boldsymbol{A}_x \boldsymbol{K}_x$, $\boldsymbol{B}_x \boldsymbol{L}_x = \boldsymbol{Q}_x^{-\top}$ and

$$\boldsymbol{C}_{x,x'} \equiv \boldsymbol{A}_{x'} \boldsymbol{P}_{x,x'} \boldsymbol{B}_x^\top = \boldsymbol{Q}_{x'} \boldsymbol{D}_{x,x'} \boldsymbol{Q}_x^{-1}.$$

   If every $\boldsymbol{Q}_x$ were known in one common column order, (3.3) would give the kernel. *Assumption used:* Assumption 1.

4. **Each $\boldsymbol{Q}_x$ up to scale and order** (Step 2, pp. 4–5). Along a walk returning to x,

$$\boldsymbol{C}_{x_m,x} \cdots \boldsymbol{C}_{x_1,x_2} \boldsymbol{C}_{x,x_1} = \boldsymbol{Q}_x (\boldsymbol{D}_{x,x_1} \cdots \boldsymbol{D}_{x_m,x}) \boldsymbol{Q}_x^{-1},$$

   whose z-th eigenvalue is the probability of that return walk for type z, $k_z^m(x; x_1, \dots, x_m; x)$, the product of its one-step kernel entries. *Assumption 2* (p. 3): for each x there is $\bar{m}$ such that the columns of $\boldsymbol{F}_x = (\boldsymbol{F}_{x,0}^\top, \dots, \boldsymbol{F}_{x,\bar{m}}^\top)^\top$, stacking these probabilities by type, are all distinct. Then the joint diagonalizer is unique up to scale and order, $\tilde{\boldsymbol{Q}}_x = \boldsymbol{Q}_x \boldsymbol{\Omega}_x \boldsymbol{\Delta}_x$ (De Lathauwer et al., 2004, Thm. 6.1, as cited; not inspected by us). The scale follows from $\boldsymbol{B}_x \boldsymbol{p}_x = \boldsymbol{Q}_x^{-\top} \boldsymbol{\iota}_q$ and Lemma A.1 (p. 10).

5. **One common order** (Step 3, (3.4), p. 5). *Assumption 3* (p. 3): there is one x_0 such that every $x \neq x_0$ is connected to or from x_0 by a walk whose probability is non-zero for each z. Then $\tilde{\boldsymbol{Q}}_x^{-1} (\boldsymbol{C}_{x_m,x} \cdots \boldsymbol{C}_{x_0,x_1}) \tilde{\boldsymbol{Q}}_{x_0}$ is a permutation times an invertible diagonal matrix; its nonzero pattern reveals $\boldsymbol{H}_{x,x_0} = \boldsymbol{\Delta}_x^{-1} \boldsymbol{\Delta}_{x_0}$, and every state takes the order $\boldsymbol{\Delta}_{x_0}$.

6. **Theorem 1** (Step 4, p. 5). The diagonal of $\bar{\boldsymbol{Q}}_{x'}^{-1}\boldsymbol{C}_{x,x'}\bar{\boldsymbol{Q}}_x$ gives the kernel; with $\boldsymbol{p}_x^* = (p_{x,1},\ldots,p_{x,r})^\top = \boldsymbol{K}_x\boldsymbol{\lambda}_x$ and $\boldsymbol{\lambda}_x = (s_1(x)\mu_1,\ldots,s_q(x)\mu_q)^\top$, the left inverse $\boldsymbol{K}_x^+$ (Assumption 1) gives $\boldsymbol{\lambda}_x$, hence μ_z and s_z. Under Assumptions 1–3, (μ_z, s_z, k_z) are identified from (X_1, X_2, X_3, X_4). Nothing requires $k_z(x,x') > 0$; the numerical example (p. 8) has absorbing states.

7. **Longer panels and higher order** (Remark and Corollary 1, p. 6; Sec. 6, p. 10). With T observations, histories are stacked in $\boldsymbol{K}_x$ and $\boldsymbol{L}_x$; at most $r^{(T-2)/2}$ types (T even) or $r^{(T-3)/2}$ (T odd); $q = \operatorname{rank}\boldsymbol{P}_x$ under Assumption 1. Order-p dependence needs $3+p$ observations under "suitable modifications" of Assumptions 1–3, which are not stated.

8. **Binary illustrations** (5.5)–(5.6), pp. 7–10. With $\omega_z = k_z(1,2)$, $\varpi_z = k_z(2,2)$, $\pi_z = s_z(1)$: $\boldsymbol{P}_1$ is invertible iff $\omega_1 \neq \omega_2$ and $\pi_1(1-\omega_1)(1-\pi_2)(1-\varpi_2) \neq \pi_2(1-\omega_2)(1-\pi_1)(1-\varpi_1)$. The second part is a restriction on initial conditions (p. 7). With binary choice, a binary state and type-invariant state transitions (attributed to Magnac and Thesmar (2002), p. 9), $\boldsymbol{K}_x$ needs only one state where type CCPs differ; the $\boldsymbol{L}_x$ part is again an initial-conditions restriction (p. 9).

9. **Comparison** (Sec. 4, pp. 6–7). Kasahara and Shimotsu (2009) (Prop. 6 in published numbering) need six observations and, applied state by state, identify $\boldsymbol{K}_x\boldsymbol{\Delta}_x$ with a different unknown permutation at each x, which does not recover kernels, initial distributions or type probabilities (p. 6). Hu and Shum (2012) need $r = q$ and monotonicity restrictions, which HJ do not (pp. 6–7). *Check:* this likely corresponds to working-paper Proposition 5; the critique targets the per-state route, not the baseline Proposition 1, which uses one global eigendecomposition.

10. **Estimator** (Appendix B, p. 11): maximum likelihood by EM on $\sum_z \mu_z \ell_z$, with type likelihood $\ell_z = \prod_x s_z(x)^{\{x_1=x\}} \prod_{x,x'} k_z(x,x')^{n_{x,x'}}$ and transition counts $n_{x,x'}$; closed-form M-step for μ_z, s_z, k_z ($r + r^2$ parameters per type), multiple starting values, outer-product-of-score standard errors. *Note:* p. 11 calls s_z "the steady-state distribution"; we read it as the initial distribution of Section 2.

Results to note. One Monte Carlo design (Tables B.1–B.2, p. 13): $r = q = 2$, $\mu_1 = 0.4$, transition probabilities 0.2–0.8, steady-state initial draws, $n = 500$, 10,000 replications. At $T = 4$, s.d. of $\hat{\mu}_1$ is 0.068 (mean 0.414); at $T = 5$, 0.046; t-test sizes 0.025–0.053. Here $|\omega_1 - \omega_2| = 0.6$, so Assumption 1 holds by a wide margin; there is no rare-event, large-r or weak-rank design. *Check:* the tables label two columns $k_2(0,1)$, $k_2(1,1)$, which the true values identify as $k_2(1,2)$, $k_2(2,2)$.

Assumptions: stated and implicit.

- *Stated by the authors:* finite, known q and $\mu_z > 0$; first-order, time-homogeneous Markov dependence given type; free initial distributions; Assumptions 1–3; random sample of complete sequences of common length (p. 11).
- *Implicit, in our reading:* "testable" means testable in the population; no rank test or near-singular analysis is given; Assumptions 1–2 must hold at every state used, which is demanding with large r and sparse cells; long walks multiply many estimated matrices; no result for time-varying kernels; no attrition; recovering utilities still needs β, the shock distribution and a reference-alternative normalization per type (Magnac and Thesmar, 2002).

Question for discussion. With deterministic vehicle age, only keepers one month younger lead into a state, so the $\boldsymbol{L}_x$ half of Assumption 1 becomes a condition on how the types arrived at each age, through other state variables or type-specific responses to them (our reconstruction); with rare purchases, is that variation detectable in samples, or would the type split rest on parametric structure?

2.2.8 Aguirregabiria et al. (2021): sufficient statistics for fixed effects in dynamic logit

Why it is included: it states which payoff parameters of a forward-looking keep-or-replace logit remain identified when persistent heterogeneity in payoffs and in the discount factor, and its dependence on the initial state, are left completely unrestricted, which shows what a random-effects type mixture identifies only through its distributional assumption.

Setup. Theoretical econometrics with a Monte Carlo and a re-estimation of Rust's bus data; $N \to \infty$, T fixed, infinite horizon (pp. 280–285). Choice $y_{it} \in \mathcal{Y} = \{0, 1, \ldots, J\}$; objective $\mathbb{E}_t[\sum_j \delta_i^j \Pi_{i,t+j}(y_{i,t+j})]$ (p. 282). *Given state:* endogenous $x_{it} = (y_{i,t-1}, d_{it})$ (lagged choice and duration in the current choice) and exogenous Markov z_{it}, observed by agent and researcher. *Unobserved by the researcher:* heterogeneity $\theta_i = (\eta_i, \delta_i)$ and shocks $\varepsilon_{it}(j)$. *Notation:* the discount factor is δ_i; β denotes payoff functions of the endogenous state. In machine replacement, keep is $y = 1$ (carries duration) and replace is $y = 0$. Method: Andersen/Chamberlain conditional maximum likelihood on a sufficient statistic for θ (FE-CML).

Math walkthrough.

1. **Payoff** (1)–(2), pp. 282–283: $\Pi_{it}(j) = \alpha(j, \eta_i, z_{it}) + \beta(j, x_{it}, z_{it}) + \varepsilon_{it}(j)$, with

$$\beta(j, x_{it}, z_{it}) = 1\{j = y_{i,t-1}\}\beta_d(j, d_{it}, z_{it}) + 1\{j \neq y_{i,t-1}\}\beta_y(j, y_{i,t-1}, z_{it}), \qquad \beta_y(j, j, z) = 0.$$

   Assumptions used: Asm. 1(B), η additively separable from x, α unrestricted, called "key" (p. 283); Asm. 1(C), $\beta_d(0, d, z) = 0$, no duration dependence in alternative 0; additive switching plus duration, "not without loss of generality" (p. 283).

2. **Transition** (3), p. 283: $d_{i,t+1} = 1\{y_{it} = y_{i,t-1}\}d_{it} + 1$ if $y_{it} > 0$, and 0 if $y_{it} = 0$. *Assumptions used:* deterministic (App. B, p. 309, adds i.i.d. jumps $\xi^d \geq 0$); Asm. 1(E), z exogenous, time-homogeneous Markov, and constant over consecutive periods (p. 285).

3. **Choice probability** (5)–(6), pp. 284–285. Since d_{t+1} is a function of (j, x_{it}),

$$P_{\theta_i}(j \mid x_{it}, z_{it}) = \frac{\exp\{\alpha(j, \eta_i, z_{it}) + \beta(j, x_{it}, z_{it}) + v_{\theta_i}(j, d_{t+1}[j, x_{it}], z_{it})\}}{\sum_{k \in \mathcal{Y}} \exp\{\alpha(k, \eta_i, z_{it}) + \beta(k, x_{it}, z_{it}) + v_{\theta_i}(k, d_{t+1}[k, x_{it}], z_{it})\}}.$$

   Assumptions used: Asm. 1(A), infinite horizon, $\delta_i \in (0, 1)$ heterogeneous; Asm. 1(D), EV1 i.i.d. over (i, t, j). Property 1 (p. 285): with $\beta_d = 0$, $v_\theta(j, z)$ does not depend on y_{t-1}. Property 2: under Asm. 2 (step 7), $v_\theta(j, d_{t+1}, z) = v_\theta(j, d_j^*, z)$ for $j = y_{t-1}$, $d_t \geq d_j^* - 1$.

4. **Binary reparametrization** (Lemma 1, (7); (11), p. 287): $y_t = 1\{\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t + 1) + \tilde{\varepsilon}_t \geq 0\}$, with $\tilde{\beta}_d(d) = \beta_d(1, d) - \beta_y(1, 0) - \beta_y(0, 1)$ and $\tilde{v}_\theta(d) = v_\theta(1, d) - v_\theta(0, 0)$. Marginal returns $\beta_d(d) - \beta_d(d-1) = \tilde{\beta}_d(d) - \tilde{\beta}_d(d-1)$, $d \geq 2$.

5. **Sufficient-statistic likelihood** (8)–(10), p. 286. For a history $\tilde{y} = (d_1, y_0; y_1, \ldots, y_T)$, if $\ln P(\tilde{y} \mid \theta, \beta) = U(\tilde{y})' g_\theta + S(\tilde{y})'\beta^*$, then

$$\ln P(\tilde{y} \mid \theta, \beta, U) = S'\beta^* - \ln\Big(\textstyle\sum_{\tilde{y}': U(\tilde{y}') = U} \exp\{S(\tilde{y}')'\beta^*\}\Big),$$

   free of θ. Lemma 2 (p. 287): β^* is identified iff there are $K + 1$ histories with the same U and a nonsingular matrix of S-differences. *Assumptions used:* Asm. 1(F), $\theta \mid (z\text{-path}, x_{i1})$ unrestricted, so $p(y_0, d_1 \mid \theta)$ is free (p. 286); EV1 for the exponential-family form. *Note:* as printed, (8) uses $\sum_{t=1}^{T}$; a history probability is a product, and (9) treats it as one.

6. **Forward-looking binary model** (20), p. 289, with $\sigma_\theta(y_{t-1}, d_t) \equiv -\ln(1+\exp\{\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t+1)\})$:

$$\ln P(\tilde{y} \mid \theta) = \ln p_\theta(y_0, d_1) + \sum_{t=1}^{T} y_t\big[\tilde{\alpha}_\theta + y_{t-1}\tilde{\beta}_d(d_t) + \tilde{v}_\theta(d_t+1)\big] + \sigma_\theta(y_{t-1}, d_t).$$

   Proposition 4 ((21), p. 290; proof (A.15), p. 305): $U = [d_1, y_0, \{H^{(1)}(d), \Delta^{(1)}(d) : d \geq 1\}]$, with $H^{(1)}(d) = \sum_t 1\{y_{t-1} = 1, d_t = d\}$ and $\Delta^{(1)}(d) = 1\{y_T = 1, d_{T+1} = d\} - 1\{y_0 = 1, d_1 = d\}$ (Table 1, p. 287). These are also the only statistics multiplying $\tilde{\beta}_d$, so *no* structural parameter is identified. The term $\tilde{v}_\theta(d_t+1)$ puts θ at every duration, and $v_\theta(1, n)$ is nonlinear in θ (fn. 19).

7. **Assumption 2 and Proposition 5** ((24), p. 290; (A.16)–(A.18), p. 305). Asm. 2 (p. 284): $\beta_d(j, d) = \beta_d(j, d_j^*)$ for $d \geq d_j^*$. By Property 2 the θ-functions are constant for $d \geq d^*$, so only sums enter U:

$$U = \Big[d_1, y_0, \{H^{(1)}(d), \Delta^{(1)}(d) : d \leq d^*-1\}, \textstyle\sum_{d\geq d^*} H^{(1)}(d), \sum_{d\geq d^*} \Delta^{(1)}(d)\Big],$$
$$S = H^{(1)}(d^*) + \Delta^{(1)}(d^*), \qquad \beta^* = \beta_d(1, d^*-1) - \beta_d(1, d^*).$$

   By Lemma 3(v), $S = X^{(1)}(d^*-1)$, the number of keeps at duration d^*-1 (our reading). *Note:* signs vary (Prop. 5, Tables 7–8); identification is unaffected. Footnote 19: "it seems" the only restrictions making $\tilde{\beta}_d(n) - \tilde{\beta}_d(n-1) + \tilde{v}_\theta(n+1) - \tilde{v}_\theta(n)$ in Example 4(b), (23), free of θ are those making it zero, a conjecture, not a theorem.

8. **Identifying** d^* (Prop. 6, (26)–(27), p. 291): for $A_n = \{0,0 \mid 1_{n-1}, 0, 1_{n+1}\}$, $B_n = \{0,0 \mid 1_n, 0, 1_n\}$, $2n+1 \leq T$, $d^* = \max\{n : \ln P(A_n) - \ln P(B_n) \neq 0\}$, provided $d^* \leq (T-1)/2$.

9. **Estimation** (43)–(48), pp. 297–298: the conditional log-likelihood $\mathcal{L}_N(\beta^*)$ is globally concave, $\sqrt{N}(\hat{\beta}^* - \beta^*) \Rightarrow N(0, J^{-1})$; d^* is chosen by $\mathrm{BIC}_N(d^*) = \ell_N(d^*) - \frac{d^*}{2}\ln N$ (consistent, Prop. 13). Enumeration cost is exponential in T (p. 297).

10. **Distribution of** θ (Sec. 3.7, (40)–(42), p. 296): payoff levels and marginal effects need a finite-support restriction or set identification. *Check:* the paper says $\tilde{v}_\theta$ can then be solved from the Bellman equation, but that requires the whole β_d function and δ_θ, which the CMLE does not identify; this step is unresolved.

Results to note. Tables 2–3 (pp. 291, 295): without duration dependence, switching costs are identified with $T \geq 3$, myopic or forward-looking (Props. 1, 2, 7, 8); myopic with duration dependence, all $\tilde{\beta}_d(d)$, $1 \leq d \leq T-2$ (Props. 3, 9); forward-looking with duration dependence, nothing in the binary model (Prop. 4); under Asm. 2, one increment per alternative (Props. 5, 11). Monte Carlo (Table 5, p. 300; $RC_i \sim N(8, 2^2)$, $\delta = 0.95$, $d^* = 3$, $N = 1000$, truth 1): CMLE 1.007/0.999/0.995 (s.d. 0.144/0.080/0.073) in samples A/B/C; a two-type MLE ignoring initial conditions 0.978/0.896/0.857; Hausman rejection at 5% 0.042/0.308/0.515, so no power with one spell per bus. Rust data (Tables 6, 8, 9): 46 of 104 buses contribute; RC_i is absorbed by $\tilde{\alpha}_\theta$ and not estimated; $d^* = 3$, $\hat{\beta}_{d^*} = 1.7009$ (1.0244) against MLE 0.36–0.45; Hausman $p = 0.19$–0.22.

Assumptions: stated and implicit.

- *Stated by the authors:* Asm. 1(A)–(F); Asm. 2 for forward-looking duration effects; stationarity, which "can be unrealistic" (p. 283); z constant within the window (p. 285); T fixed, $N \to \infty$, bounded parameter spaces (p. 297).

- *Implicit, in our reading:* flattening serves as an identifying device, and since BIC selects $d^* \leq (T-1)/2$, a short panel can only report a small d^*; the effective sample is the contributing units, which in our toy enumeration need two observed renewals (shortest $T = 2d^* + 1$ from a fresh start; pattern checked for $d^* \leq 5$, not proved); Monte Carlo replacement rates and contributing shares are not reported; Sec. 3.7 omits the unidentified parts of β_d and δ.

Question for discussion. In a monthly household keep/replace model whose replace payoff does not depend on vehicle age, the only flattening point is the state cap on age: is there an economic reason for a flat age cost after a few years, or would FE-CML identification there be an artifact of truncation?

2.3 C3. Latent household state

2.3.1 Li et al. (2011): latent financial states and bank cross-selling

We work from the authors' November 2010 manuscript (third revision), not the published article, so we cite equations, tables, and sections; page numbers are manuscript pages, and "A" marks appendix equations. *Why it is included:* it is a worked template, in bank data, for a latent time-varying household state with a covariate-dependent transition equation and a state-dependent measurement equation for monthly product openings.

Setup. One bank, 2,000 calibration households, monthly, November 2003 to January 2006 (27 months), seven product categories (Sec. 3). *Given state:* a latent "financial state" $s \in \{1, \ldots, S\}$, occupied by the household but observed by neither the bank nor the researcher; observed covariates (balance changes, accounts, transactions, tenure, income). *Actions:* household openings Y_{ijt}; bank solicitations Z_{ijkt} for product j by channel k (mail, email), observed by the researcher. *Objectives:* the household maximizes current utility only; the bank maximizes discounted profit (Sec. 5). The state is measured only through openings.

Math walkthrough.

1. **Latent utility** (1)–(2). $Y_{ijt} = 1$ if i opens j in month t, with

$$\begin{aligned} U_{ijt}(s) = {} & \beta_{0ij}(s) + \textstyle\sum_k (\beta_{1ij}(s) + \beta_{2ik}(s)) Z_{ijkt} + \beta_{3i}(s) \sum_k \sum_j \sum_{\tau=1}^{t-1} Z_{ijk\tau} \\ & + \beta_{4i}(s) \mu_{\Delta BAL_{i(t-1)}} + \beta_{5i}(s) \sigma_{\Delta BAL_{i(t-1)}} + \beta_{6i}(s) NACCT_{ijt} + \beta_{7i}(s) TRANS_{i(t-1)} \\ & + \cdots + \beta_{10i}(s) INCOME_{it} + \varepsilon_{ijt}(s), \end{aligned}$$

   where the omitted terms are $TENURE_{it}$ and $COMP_{it}$. *Assumptions used:* every coefficient is state-specific; lagged balance-change and transaction covariates are exogenous to solicitations (fn. 3); a solicitation shifts current utility and the cumulative stock shifts later utility. *Note:* (2) has no continuation value, so the household is myopic given its state.

2. **Orthant probability** (3)–(4). $\boldsymbol{\varepsilon}_{it}(s) \sim MVN[0, \Sigma]$ across the J products, and $\mathbb{P}(\mathbf{Y}_{it} \mid s) = \int_{M_j} \phi_J(\boldsymbol{\varepsilon}_{it}(s); 0, \Sigma)\, d\boldsymbol{\varepsilon}_{it}(s)$, with $M_j = (-\infty, 0)$ if $Y_{ijt} = 0$ and $(0, \infty)$ otherwise, writing ϕ_J for the printed normal density. *Assumptions used:* $Y_{ijt} = \mathbf{1}\{U_{ijt}(s) > 0\}$; errors independent over time given s; Σ has unit diagonal (App. Sec. 2). *Check:* with $U = \bar{U} + \varepsilon$, $Y_{ijt} = 1$ iff $\varepsilon_{ijt} > -\bar{U}_{ijt}(s)$, so the printed M_j give the sign of U and must be shifted by $-\bar{U}_{ijt}(s)$ to apply to ε. For $J = 1$, (4) is $\Phi(\bar{U}_{i1t}(s))$ or its complement; unconditionally, $\mathbb{P}(\mathbf{Y}_{it}) = \sum_s \mathbb{P}_{it}(s) \mathbb{P}(\mathbf{Y}_{it} \mid s)$.

3. **Transition matrix** (5). M_{it} is $S \times S$ with zero diagonal; P_{itmn} is the probability of a jump from m to n. *Assumptions used:* first-order switching among states common to all households;

persistence comes from waiting times. *Note:* a discrete-time version with self-transitions gave "very similar results" (fn. 4), not shown.

4. **Gamma waiting times** (6)–(7). $W_{it}(s) \sim$ Gamma with shape $\lambda_{it}(s)$ and inverse scale $k_i(s)$, so $E[W_{it}(s)] = \lambda_{it}(s)/k_i(s)$. *Check (hazard):* $h(w) = f(w)/\{1 - F(w)\}$. For $\lambda = 1$, $f = ke^{-kw}$ and $1 - F = e^{-kw}$, so $h = k$, with no duration dependence; for $\lambda > 1$ the hazard rises with w, for $\lambda < 1$ it falls, both toward k. A monthly Markov chain with self-transitions forces geometric durations, so duration dependence requires the semi-Markov form (state plus elapsed time).

5. **Covariate-dependent shape** (8)–(9). $\log \lambda_{it}(s) \sim N(\bar{\lambda}_{it}(s), \sigma^2_\lambda)$, with $\bar{\lambda}_{it}(s)$ linear in experience $TACCT_{i(t-1)}$, current solicitations ($\alpha_{2ij}(s)$), the cumulative channel stock and its square, $COMP$, $INCOME$, and closed accounts. *Assumptions used:* only the shape moves; $k_i(s)$ does not. $\alpha_{2ij}(s) < 0$ shortens the stay in s, the "educational" channel the authors call speculative (p. 13). Past purchases enter through $TACCT$, so the state depends on the household's own choices. *Note:* the text resets the stock at entry to the current state; the printed sum starts at $t_s = 1$.

6. **Priors and initial state** (10)–(12). Transition rows and initial probabilities are Dirichlet with log-normal hyperparameters, and $\bar{\eta}_{is} = \omega_{0i} + \omega_{1i} TACCT_{i0} + \omega_{2i} ASSET_{i0}$. *Note:* (12) is a parametric initial-condition equation in start-of-sample holdings; elapsed time in the first spell (left censoring) is not treated.

7. **Hierarchy** (13). $\theta_{im} = \mu_{m0} + \mu_{m1} AGE_i + \mu_{m2} GENDER_i + \mu_{m3} HSIZE_i + e_{im}$, $e_i \sim N[0, \Omega]$, for the stacked utility parameters. The heterogeneity distribution of $\alpha_i(s)$, $k_i(s)$, ω_i is not specified.

8. **Endogenous solicitations** (A1)–(A2). $Z_{ijkt} \mid \gamma_i \sim$ Poisson(γ_i), with $\ln \gamma_i$ linear in state-averaged coefficients on $NACCT$, $TENURE$, $INCOME$ and averaged hierarchy coefficients. *Assumptions used:* the bank knows average response parameters but not the current state, so Z is independent of s and ε given Θ_i. *Check:* the last three terms of (A2) average population hyperparameters and cannot be separated from φ_0; (A1) treats Z as a count, elsewhere it is binary.

9. **Likelihood** (A3)–(A4). For the realized latent path D_{it},

$$\mathbb{P}(\mathbf{Y}, \mathbf{Z}) = \prod_i \prod_s (\pi_i(s))^{\mathbf{1}\{D_{i0}=s\}} \Big\{ \prod_t \mathbb{P}(Y_{it} \mid D_{it}, Z_{it}) \mathbb{P}(Z_{it}) \cdot \Big(\prod_l P_{itsl}^{\mathbf{1}\{D_{it}=s \,\&\, D_{i(t+1)}=l\}} \Big) \prod_k f(W_{itk} \mid D_{it}, \lambda_{it}, \kappa_i) \Big\},$$

where (A4) is a gamma density restricted to the interval between jump times τ_{k+m} and τ_{k+m+1}. *Assumptions used:* complete-data likelihood with MCMC data augmentation. *Note:* samplers, burn-in, and convergence checks are not reported; (A4) writes the rate κ_i, (6) writes $k_i(s)$.

10. **Ordering and number of states** (A5), Table 3A. With mean-centered covariates,

$$\tfrac{1}{J} \textstyle\sum_j \beta_{0ij}(1) \le \tfrac{1}{J} \sum_j \beta_{0ij}(2) \le \cdots \le \tfrac{1}{J} \sum_j \beta_{0ij}(S).$$

This fixes labels, not S or the transition law. Log marginal densities for $S = 1, \ldots, 4$ are $-1{,}154{,}047.7$, $-1{,}144{,}416.0$, $-984{,}746.9$, $-1{,}184{,}113.2$, so $S = 3$. *Check:* the sequence is not monotone, and its computation is not described.

11. **Bank dynamic program** (15)–(18). Expected profit (15) weights state-specific purchase probabilities by filtered $\mathbb{P}_{it}(s)$, margins r_{ij}, less channel costs c_k; (16) sums it over the 27 months; (17) adds an extreme-value shock, giving the logit policy (18). *Note:* the DP is solved with estimated customer parameters, not estimated from bank behavior. As printed, $(1+\delta)^{t-\xi_1}$ in (16) compounds rather than discounts, and δ is not reported.

Results to note. $S = 3$ fits best (Table 3A); the full model has the best validation purchase hit rate, 0.268 against 0.101–0.246 (Table 3B). Mean balance change raises utility (1.06 to 1.29) and its dispersion lowers it (-1.45 to -1.92) (Table 4A). Mean waiting times are 9.68, 10.90, 15.07 (Table 4D; units presumably months), so 27 months cover roughly two or three spells, the first left-censored. The simulated dynamic policy raises profit from 22,021.5 to 60,925.5 (Table 6) under the same fitted model.

Assumptions: stated and implicit.

- *Stated by the authors:* S common states; zero-diagonal semi-Markov switching with gamma durations; MVN errors, unit-diagonal Σ; solicitations exogenous given average parameters (A1)–(A2); balances exogenous (fn. 3); ordering (A5); finite-horizon bank problem.
- *Implicit, in our reading:* myopic customers; states identified from rare openings (about 3.1% of observations, p. 19), with persistent heterogeneity (13) and time-varying states competing for the same serial correlation (Heckman, 1981); balances shift utility rather than signal the state; no bank targeting on the current state.
- *Reporting problems:* Table 2 prints a mean balance change (-6.35) outside its range (-0.033 to 3.8), $TACCT$ up to 155 with seven categories, and tenure in years with mean 66.46. Table 4A's waiting-time intercepts (-8.23, -8.53, 13.43) are not consistent log shapes for Table 4D; combining Table 4D with Table 4A's $k_i(s)$ through (7) gives shapes 61.66, 54.94, 16.88 (our computation), nearly deterministic durations.

Question about the paper.

- With openings as the only measurement, what separates the latent financial state from the persistent heterogeneity in (13) over 27 months?

2.4 C4. Liquidity and purchase

2.4.1 Adams et al. (2009): liquidity constraints in subprime auto purchases

Why it is included: it is direct evidence that the up-front cash outlay, separately from the financed price, governs purchase: subprime used-car applicants respond strongly to the required down payment and to tax-refund season, and barely to the financed price.

Setup. Unit: a loan application at one large subprime dealer-lender, June 2001 to December 2004 (p. 52); over 50,000 applications, 34 percent purchase (Table 1). *Outcome:* $q_i = 1$ if applicant i buys, given that i applied. *Set by the company:* minimum down payment d_i (by credit category and region) and list price l_i (a markup over cost). *Negotiated:* price p_i. *Chosen by the buyer:* buy or not, and $D \geq d$. Assets and debt are not observed (p. 53).

Math walkthrough.

1. **Purchase probit** (1), p. 59: $q_i = 1 \Leftrightarrow q_i^* = \mathbf{x}_i'\boldsymbol{\alpha} + \varepsilon_i \geq 0$, where $\mathbf{x}_i$ holds offer terms (price, rate, term, minimum down), car and applicant characteristics, and city, credit-category, month and year dummies. *Assumptions used:* ε_i i.i.d. normal; one binary decision on the lot; terms affect neither who applies (they are not posted) nor car choice (p. 60).

2. **Missing prices** (p. 59, fn. 17). *Assumption used:* a nonpurchaser faced the car and price of a random purchaser in the same city, week, credit and income category. *Note:* prices are imputed for about two thirds of the sample, which attenuates the price effect; fn. 17 notes a further bias toward underestimating it.

3. **Minimum-down identification** (p. 60). Given the dummies, the effect comes from company changes over time and differential changes across categories and cities. *Assumption used:* company decisions are uncorrelated with ε_i given controls, since analysts see what headquarters sees. *Implicit, in our reading:* minimums are not timed to demand shifts the month dummies miss (they are about \$150 higher in February, p. 57).

4. **Negotiated-price equation** (2), p. 60: $p_i = l_i\lambda_l + \mathbf{x}_i'\boldsymbol{\lambda}_x + \nu_i$, estimated jointly with (1) by maximum likelihood, (ε_i, ν_i) correlated normal. *Assumptions used:* l_i is excluded from (1) given cost controls; it varies through changes in the margin schedule, regions, and discontinuities of the margin formula in cost. Relevance is strong ($t > 100$); exclusion is not testable. Columns 3–4 use cost-bucket and state dummies instead (fn. 18).

5. **Expected payment without a liquidity constraint** (3), p. 61:

$$\mathbb{E}[Payment] = D + (p - D)\,\frac{\sum_{t=1}^{T}(1+s)^{-t}S_t}{\sum_{t=1}^{T}(1+r)^{-t}} = D + (p - D)\phi,$$

   with s the buyer's monthly discount rate, S_t the probability of no default before the end of month t, and ϕ the expected present value paid per dollar borrowed; the denominator is the annuity factor (fn. 7). *Assumption used:* an unconstrained buyer at the minimum ranks offers by $\mathbb{E}[Payment]$. *Reconstructed step:* ΔD changes it by $\Delta D(1-\phi)$ and Δp by $\Delta p\,\phi$, so indifference requires $\phi = \Delta D/(\Delta D + \Delta p)$. *Check:* \$100 of down payment worth about \$30 of price at 5 percent (p. 61) implies $\phi \approx 0.77$.

6. **Coefficients as a liquidity test** (p. 63). The ratio of the minimum-down to the price effect is the empirical Δp per \$100 of ΔD: \$2,884, so $\phi \approx 0.034$, which (3) matches only at a 1,415 percent annual discount rate (474 percent with col. 4); at 10, 20 and 50 percent the benchmarks are \$38, \$55, \$108. *Note:* the col. 2 price effect, −0.0010 (0.0011), is not distinguishable from zero, so the ratio is bounded below rather than estimated, and neither \$2,884 nor 1,415 percent has a reported standard error. The rounded coefficients give about \$2,990. S_t and the compounding convention are not reported (fn. 19), so we read 1,415 percent as an order of magnitude.

7. **Cell-level demand** (4), p. 64: $\ln Q_j = \beta_A \ln A_j + \tilde{\mathbf{x}}_j'\boldsymbol{\beta}_x + u_j$, cells = city × week × credit category × income range; OLS (col. 5) and IV with average list price (col. 6). It avoids the imputation of step 2; dividing by three roughly converts to the probit scale (Table 2, note d).

8. **February evidence** (pp. 56–59). Applications are 23 percent higher and the close rate is 40 versus 33 percent, almost 50 percent more sales (p. 50). Across 12 income-by-dependents groups the February increase rises with the group's earned income tax credit (Fig. 2B); trade-in purchases show no seasonality. *Assumption used:* the spike reflects tax refunds. *Note:* a descriptive gradient without standard errors; whether February buyers would otherwise buy later is not observed.

The default, loan-size and adverse-selection analyses, (5)–(9), are outside our current scope.

Results to note (Table 2, col. 2, dF/dx, per \$100 where applicable). Minimum down −0.0299 (0.0006), about 9 percent of the 0.34 purchase rate; price −0.0010 (0.0011); February 0.1594 (0.0047) relative to December. Robustness subsamples give −0.0238 to −0.0304 (p. 65); cell level −0.0895

(0.0039) log points (col. 5). Standard errors use 50 bootstrap replications. *Reconciliation:* the text puts February demand 30 percent higher (p. 59), but 0.1594/0.34 is larger; text and Fig. 2B differ on the top income group ($3,500 versus $3,000).

Assumptions: stated and implicit.

- *Stated by the authors:* normal errors; one binary decision; minimums exogenous given dummies; list price a valid instrument; imputed nonpurchaser prices; no effect of terms on applications or car choice.
- *Implicit, in our reading:* purchase is one-shot, not a choice of when to buy; impatience or present bias is set aside by plausibility, not a test; the dealer's applicants represent the population of interest.

Question about the paper.

- Given the imprecise price effect, do we cite $2,884 as a lower bound, or only the down-payment effect?

2.5 C5. Initial conditions and baseline

2.5.1 Heckman (1981): initial conditions and incidental parameters

Why it is included: it supplies the likelihood term needed when units enter the data partway through their histories and differ in a persistent way the analyst does not observe: a model for the first observed state given the household's type.

Setup. Individuals $i = 1, \dots, I$ follow a binary first-order Markov process over $t = 1, \dots, T$ that starts at a fixed $d(i, 0)$, but the analyst observes it only from period $J > 0$ onward (pp. 179, 183). *Observed:* $d(i, J), \dots, d(i, T)$, and in extensions covariates $\mathbf{Z}(i, t)$. *Unobserved effect:* a persistent $\tau(i)$ with density $f(\tau)$. No decision problem is specified.

Math walkthrough.

1. **Latent index model** (4.1), p. 181:

$$Y(i,t) = \beta_0 + \gamma d(i, t-1) + \tau(i) + U(i,t), \qquad d(i,t) = 1 \text{ iff } Y(i,t) \geq 0.$$

   γ is state dependence; $\tau(i)$ makes the composite error serially correlated. *Assumptions used:* $E[U] = E[\tau] = 0$, $\sigma_U^2 = 1$, $E[\tau^2] = \sigma_\tau^2$; U serially independent and independent of τ; $d(i, 0)$ fixed and nonstochastic (p. 181).

2. **Transition given** τ (4.2), p. 181:

$$\mathbb{P}[d(i,t) \mid d(i,t-1), \tau(i)] = \Phi\{[\beta_0 + \gamma d(i,t-1) + \tau(i)][2d(i,t) - 1]\}.$$

   Assumptions used: U normal, which the author calls "inessential" (p. 181).

3. **Probability of the first observed state.** Equation (4.3), p. 182, sums the probabilities of all paths from $d(i, 0)$ to $d(i, J)$, so $P[d(i, J) \mid \tau]$ depends on β_0, γ, $d(i, 0)$ and τ (p. 184). In the stationary case, (4.4), p. 182,

$$\Pi_1(\tau) = \frac{\Phi(\beta_0 + \tau)}{1 - \Phi(\beta_0 + \gamma + \tau) + \Phi(\beta_0 + \tau)}, \qquad \Pi_0 = 1 - \Pi_1.$$

Assumptions used: bounded β_0, γ, τ (ergodicity) and equilibrium by period J. *Check* (from fn. 1): stationarity $\Pi_1 = p_{11}\Pi_1 + p_{01}(1 - \Pi_1)$ with $p_{01} = \Phi(\beta_0 + \tau)$, $p_{11} = \Phi(\beta_0 + \gamma + \tau)$ gives (4.4), which is not a probit (p. 182). *Note:* (4.3) is printed as a product of sums; we read it as nested sums over paths.

4. **Correct likelihood** (4.7), p. 183, for a sample starting at J:

$$\mathcal{L} = \prod_{i=1}^{I} \int_{-\infty}^{\infty} \prod_{t=J+1}^{T} \Phi\{[\beta_0 + \gamma d(i, t-1) + \tau][2d(i,t) - 1]\}\, P[d(i,J) \mid \tau]\, f(\tau)\, d\tau.$$

   The conditional version (4.6) divides by $\int P[d(i,J) \mid \tau] f(\tau)\, d\tau$. Either is consistent as $I \to \infty$ for $T \geq 2$ (p. 184). *Assumptions used:* parametric f (normal "for convenience," p. 182).

5. **Naive likelihood** (4.8), p. 184: the same expression with $P[d(i,J) \mid \tau]$ omitted, treating $d(i,J)$ as exogenous. It is inconsistent for $\beta_0, \gamma, \sigma_\tau^2$ (p. 184); the proof is only sketched (fn. 3). *Check* (our reconstruction): by Bayes' rule $P[d(i,J) \mid \tau] f(\tau) \propto f(\tau \mid d(i,J))$, so (4.6) integrates over τ *given* the first state and (4.8) over $f(\tau)$. They agree only if $P[d(i,J) \mid \tau]$ is flat in τ: degenerate f, or a new process at J (p. 179). Those starting in state 1 have higher τ, and (4.8) credits that persistence to γ.

6. **Exact solutions and their data requirements** (pp. 183–185). (a) The full history from $d(i,0)$, likelihood (4.5), p. 182, is "hard to come by" (p. 184). (b) Initial stationarity, (4.4) in (4.7), is "computationally forbidding" (p. 184) and, with $\mathbf{Z}(i,t)\boldsymbol{\beta}$ in place of β_0, needs a stationary $\mathbf{Z}$ process, which "excludes time and age trends" (p. 184). (c) Presample $\mathbf{Z}$ can be integrated over a finite-parameter density g estimated elsewhere; plugging in means is not correct, because $\mathbf{Z}$ enters nonlinearly (p. 185).

7. **Fixed effect likelihood** (4.9), p. 186. Treating each $\tau(i)$ as a parameter, $P[d(i,J) \mid \tau(i)]$ cancels from (4.6):

$$\mathcal{L} = \prod_{i=1}^{I} \prod_{t=J+1}^{T} \Phi\{[\beta_0 + \gamma d(i, t-1) + \tau(i)][2d(i,t) - 1]\}.$$

   It needs no presample data and no f, but stayers give $\hat{\tau}(i) = \pm\infty$ and drop out, and for fixed T the inconsistent $\hat{\tau}(i)$ contaminate $\hat{\gamma}, \hat{\beta}$: the incidental-parameters problem (pp. 186–187). *Check* (fn. 9, p. 187): with $\gamma = 0$, $T = 2$ and a period effect β, movers give $\hat{\tau}(i) = -\beta/2$ and, as $\sigma_\tau^2 \to 0$, $\operatorname{plim} \hat{\beta} = 2\beta$.

8. **Approximate solution** (pp. 188–189). Model the first state by a reduced-form probit, $Y(i,J) = \sum_{l=0}^{J} D(l)[\mathbf{Z}(i,l)] + \mu(i,J)$, $D(l)$ polynomials in practice (fn. 11), with $\mu(i,J)$ freely correlated with $\tau(i) + U(i,t)$, $t > J$, and no cross-restrictions to the structural parameters. *Assumptions used:* consistency for $I \to \infty$, fixed T, holds "assuming that the model is exact" (p. 189). *Note:* by step 3 the true marginal is not a probit, so in our reading consistency is not guaranteed; the correlation's parameterization is not stated.

Results to note. Monte Carlo, 25 replications of $I = 100$, no dispersion reported (pp. 189–193). Without a lagged state (Table 4.1, p. 191), the fixed effect probit at $T = 8$ is biased at most 10%, toward zero; the third design reads $\beta = -0.1$ in the table header but $\beta = 0.1$ in the text (p. 189), and the entries $(-0.10, -0.09, -0.10)$ fit -0.1. With a lagged state (Table 4.2, p. 193), fixed effects at $T = 8$ give $\hat{\gamma}$ of 0.03 to 0.25 for true 0.5 and the wrong sign in all six cells for true 0.1; the approximate estimator at $T = 3$ gives 0.54 to 0.70 for 0.5. All estimators are biased at $I = 100$,

$T = 3$ (p. 194). *Illustration with invented parameters, not evidence* (seed 20261023, $I = 1000$, 100 replications, true $\gamma = 0.5$; mean $\hat{\gamma}$, standard deviation in parentheses): with 3 observed periods, an exogenous first state gives 1.350 (0.077), the stationary correction 0.507 (0.172), the approximate initial probit 0.503 (0.195); with 8 periods, 0.632 (0.056) versus 0.502 (0.049) for both corrections. With true $\gamma = 0$ and 8 periods, a pooled probit gives 0.880 (0.046).

Assumptions: stated and implicit.

- *Stated by the author:* first-order Markov binary state; $\tau + U$ components of variance with U serially independent and independent of τ; normal U and, for convenience, normal f; bounded exogenous $\mathbf{Z}$ (p. 184); exactness of the approximate initial probit (p. 189).
- *Implicit, in our reading:* τ independent of $\mathbf{Z}$; independence across individuals; the same stable process before and after J; no forward-looking behavior.

2.5.2 Keane and Wolpin (1997) and Card and Hyslop (2005): latent types with initial conditions

We work from the published article for Keane and Wolpin (journal pages) and from NBER Working Paper 10647 for Card and Hyslop, not the published Econometrica article, so Card–Hyslop page numbers are working-paper pages, tables are cited by number, and the published equations and estimates may differ. *Why it is included:* the two papers handle the state in which persistent types are first observed in two different ways, type shares conditional on the initial state and a sample in which everyone shares the same initial state, and a household first observed partway through owning a vehicle needs one of them.

(a) Keane and Wolpin: type shares conditional on initial schooling.

Setup. 1,373 white males, NLSY79 core sample, annual from age 16 through September 1988 (p. 483); five exclusive choices: white-collar, blue-collar, military, school, home (p. 477); terminal age $A = 65$ (p. 490). Finite-horizon structural model, simulated maximum likelihood with an approximated EMAX (p. 481; App. A). *Given state:* $\mathbf{S}(a) = \{\mathbf{e}(16), g(a), \mathbf{x}(a), \boldsymbol{\epsilon}(a)\}$, with schooling g and experience $\mathbf{x}$ observed by both, the endowment $\mathbf{e}(16)$ and shocks only by the agent. Wages are observed in chosen occupations.

Math walkthrough.

1. **Rewards** (3), p. 479: $w_m(a) = r_m \exp[e_m(16) + e_{m1} g(a) + e_{m2} x_m(a) - e_{m3} x_m^2(a) + \epsilon_m(a)]$, $m = 1, 2, 3$; $R_4(a) = e_4(16) - tc_1 I[g(a) \geq 12] - tc_2 I[g(a) \geq 16] + \epsilon_4(a)$; $R_5(a) = e_5(16) + \epsilon_5(a)$. The extended model adds mobility, re-entry, diploma and age terms ((C1)–(C3), pp. 520–521). *Assumptions used:* $\boldsymbol{\epsilon}(a) \sim N(0, \Omega)$, serially uncorrelated (p. 479, fn. 9).
2. **Dynamic program** (5)–(6), p. 480: $V_m(\mathbf{S}(a), a) = R_m(\mathbf{S}(a), a) + \delta E[V(\mathbf{S}(a+1), a+1) \mid \mathbf{S}(a), d_m(a) = 1]$ for $a < A$, $V_m = R_m$ at A, $V = \max_m V_m$. *Assumptions used:* deterministic accumulation $x_m(a+1) = x_m(a) + d_m(a)$, $g(a+1) = g(a) + d_4(a)$ (p. 480).
3. **Likelihood without types** (7), p. 481, with $\bar{\mathbf{S}}(a)$ the state net of shocks and $c(a)$ the choice and, when working, the wage: $\mathbb{P}[c(16), \ldots, c(\bar{a}) \mid g(16), \mathbf{e}(16)] = \prod_{a=16}^{\bar{a}} \mathbb{P}[c(a) \mid \bar{\mathbf{S}}(a)]$. *Assumption used:* serial independence of shocks.
4. **Types** (pp. 481–482, 490): $K = 4$, each an endowment vector $\mathbf{e}_k(16) = \{e_{mk}(16) : m = 1, \ldots, 5\}$ of intercepts. If $g(16)$ were exogenous, the likelihood would be $\prod_n \sum_k \pi_k L_{nk}$ (p. 482). *Assumptions used:* K known; technology, δ and Ω common to types; no military endowment heterogeneity (fn. 26).

5. **Initial-condition device** (8), p. 482. KW "assume that initial schooling is exogenous conditional on the age 16 endowment vector" (p. 482), and the intended contribution is

$$\mathbb{P}[c_n(16), \ldots, c_n(\bar{a}) \mid g_n(16)] = \sum_{k=1}^{K} \pi_{k|g_n(16)} \prod_{a=16}^{\bar{a}} \mathbb{P}[c_n(a) \mid g_n(16), \text{type} = k].$$

Note: as printed, $\pi_{k|g}$ sits inside the product, a typesetting slip given the surrounding text. $\pi_{k|g}$ are free in two cells, $g(16) \in \{7, 8, 9\}$ and $\{10, 11\}$ (fn. 26, p. 490); $x_m(16) = 0$ (p. 479). Footnote 13 (p. 482): if data began at $a_0 > 16$, the likelihood would condition on $g(a_0)$ and $x_m(a_0)$, with type probabilities "estimated separately for each state combination."

6. **Equivalence with Heckman's conditional likelihood** (our reconstruction; not stated by KW). Bayes' rule gives $\mathbb{P}[c_n \mid g_n(16)] = \sum_k \pi_{k|g} L_{nk}$ with $\pi_{k|g} \equiv \pi_k \mathbb{P}[g \mid k] / \sum_{k'} \pi_{k'} \mathbb{P}[g \mid k']$, the discrete-type form of Heckman's (4.6) (Section 2.5.1). With $\mathbb{P}[g \mid k]$ unrestricted, any free $\{\pi_{k|g}\}$ is generated by some $(\pi_k, \mathbb{P}[g \mid k])$, so (8) is exact, not an approximation like Heckman's probit reduced form. *Assumptions used:* exogeneity of $g(16)$ given type and serial independence (to write $\mathbb{P}[c_n \mid g_n(16), k] = L_{nk}$); saturation in the initial state. With many initial-state cells, pooling or parameterizing $\pi_{k|s}$ reintroduces approximation.

7. **Identification.** No argument is given for types, shares, or $\pi_{k|g}$. Types are disciplined by parametric structure and by wages, whose intercepts are the endowments (eq. (2), p. 478; App. B).

Results to note. Table 9 (p. 502, extended model): $\pi_{k|g} = (.0491, .1987, .4066, .3456)$ for nine years or less, $(.2343, .2335, .3734, .1588)$ for ten or more. In the basic model (Table B1 panel C) type rankings differ, so labels are not comparable across specifications. Between-type variance is 90% of the variance of expected lifetime utility (p. 508); family background explains under 10% (Table 13). Within-sample χ^2 (Table 5, p. 495; not adjusted for estimated parameters): basic model rejects strongly, extended fits at most ages. Out of sample (Table 10, p. 505), white-collar shares track the CPS (.388 vs .413 at 30–33); blue-collar is overstated (.547 vs .460). Not shown: any estimate without the device, a test of exogeneity given type, or sensitivity to K.

Assumptions: stated and implicit.

- *Stated by the authors:* $K = 4$ known; initial schooling exogenous given the endowment; serially independent normal shocks; types shift intercepts only; free $\pi_{k|g}$ in two cells.
- *Implicit, in our reading:* any persistent unobservable that drove $g(16)$ other than the type is ruled out; a counterfactual that changes the distribution of $g(16)$ is outside the model; wage data supply identifying information that a choice-only panel lacks.

(b) Card and Hyslop: a common initial state by sample design.

Setup. Self-Sufficiency Project experiment, British Columbia and New Brunswick (p. 4): single-parent Income Assistance (IA) recipients on welfare in at least 11 of the previous 12 months, randomized; monthly IA records for 69 months (p. 6). Program members could receive a three-year earnings subsidy only if they began full-time work within 12 months (p. 4). A search model (pp. 16–19) motivates but is not estimated. *Outcome:* $y_{it} = 1$ if on IA. *Unobserved:* α_i.

Math walkthrough.

1. **Sample restriction** (p. 6). The 5,617 people on IA in *both* months before random assignment. "This restriction eliminates any 'initial conditions' problems associated with the second order dynamic models" (fn. 10); with $y_{i0} = y_{i,-1} = 1$ for all, the distribution of initial conditions need not be modeled (fn. 28, p. 21).

2. **Dynamic logit with random effect** (1), p. 21, with x_{it} a fourth-order polynomial in months since assignment:

$$P(y_{i1},\ldots,y_{iT} \mid x_{i1},\ldots,x_{iT}) = \int \Big\{\prod_t L(\alpha_i + x_{it}\beta + \gamma_1 y_{it-1} + \gamma_2 y_{it-2} + \gamma_3 y_{it-1}y_{it-2})\Big\} f(\alpha_i)\, d\alpha_i.$$

f is $N(0,\sigma_\alpha^2)$ (10-point quadrature, fn. 29) or discrete with estimated mass points (p. 21). Preferred control specifications let the state-dependence terms vary linearly with α_i (Table 4, cols. 3–4). *Assumptions used:* second-order Markov given α_i; no baseline covariates (fn. 31).

3. **Why the restriction works** (our reconstruction). Heckman's conditional likelihood with $s_{i0} = (y_{i0}, y_{i,-1})$ is $\int \prod_t L(\cdot) f(\alpha \mid s_{i0})\, d\alpha$. If $s_{i0} = (1,1)$ for everyone, f in (1) *is* $f(\alpha \mid s_0 = (1,1))$: the problem is absorbed into the definition of the population, the stock of long-term recipients (fn. 8). *Assumptions used:* the conditioning set equals the full Markov state; a third-order model or baseline covariates would require $f(\alpha \mid s_0, x_{i0})$.

4. **Randomization** (pp. 16, 27): the "crucial" maintained assumption is that f is the same in both arms; it, not an initial-state model, identifies treatment effects. The restriction excluded 40 program and 27 other members, arm difference $p = 0.10$ (fn. 10).

5. **Program group** (2)–(5), pp. 27–28. IA and eligibility E_{it} load on the same α_i; (3) assumes eligibility independent of IA history given α_i. Eligibility hazard $\Phi[d(t) - k(\alpha_i)]$ while not yet eligible and $t \le T_e = 14$, with $k(\alpha) = k_0\alpha$ or one value per mass point. Treatment enters the logit index through

$$\begin{aligned}\tau(t, E_{it}, t_i^e, y_{it-1}) = {} & E_{it}1(t_i^e \le t \le t_i^e + J - 1)\{\psi_0 1(y_{it-1}=0) + \psi_1 1(y_{it-1}=1)\} \\ & + E_{it}1(t_i^e + J \le t \le t_i^e + 35)\{\lambda_0 1(y_{it-1}=0) + \lambda_1 1(y_{it-1}=1)\}, \qquad J = 3.\end{aligned}$$

Entitlement effects are "directly identified" through λ_0, λ_1; establishment effects are not (p. 29).

Results to note. With three or more mass points, one goes to the boundary (value 16 or more), "pure stayers" (p. 24); five or six add little over four. Mass-point locations and probabilities are not reported in Tables 4 or 6. $k_0 = 0.21, 0.21, 0.29$ (Table 6, cols. 2–4); median α is -0.98 among eligibles and 0.36 among ineligibles (p. 30); treating eligibility as exogenous inflates entitlement effects by about 30% (p. 30). Control fit (informal Pearson over 50 history cells, fn. 33): 752.6 (first order), 260.3 (second), 253.0 (interactions), 175.8 (four mass points) (Table 4). Over 99% of the variance of mean monthly IA participation is explained (p. 33); the never-leaver share is mispredicted (p. 32). No out-of-sample forecast.

Assumptions: stated and implicit.

- *Stated by the authors:* common initial state by design; second-order Markov given α_i; f normal or discrete and common across arms; eligibility independent of IA history given α_i.
- *Implicit, in our reading:* the estimated f describes the sampled stock only, not welfare entrants; the device needs a conditioning event fixed before any outcome is recorded; any type-dependent event used for conditioning must be modeled jointly with the type, as (2)–(5) do for eligibility.

Comparison of the two devices (our reconstruction). Both are Heckman's conditional likelihood with an unrestricted initial-state distribution; CH's is KW's with one initial-state cell, so $\pi_{k|s_0} = \pi_k$ for the sampled population. KW's validity rests on an untested behavioral assumption (exogeneity given type) and applies to initial states that vary across people; CH's rests on the sampling rule

and the Markov order, with randomization supplying arm comparability. Neither paper gives an identification argument for the mixture.

Question for discussion. For households first seen at vehicle age a and liquidity ℓ, a KW-style $\pi_\tau(a, \ell)$ needs panel entry exogenous given type and state: does a household enter the bank data independently of its vehicle decisions, or for example when it takes an auto loan, which would condition on a type-dependent purchase?

2.5.3 Shumway (2001)

TODO: pending full text. Discrete-time hazard estimated as a multi-period logit, the finance-standard baseline. To verify: the equivalence result and the out-of-sample evaluation. Supplies the reduced-form baseline, a horizon probability $1 - \prod_r \{1 - \lambda(r)\}$ built from period hazards $\lambda(r)$, that a structural model must beat.

2.6 Compatibility summary

Combination	Verdict	Reason
Rust + Schiraldi	Compatible structure	Same EV1 / CI engine; Schiraldi's aggregate identification does not transfer
Rust/Schiraldi + Gavazza–Lanteri	Needs assumptions	Add EV1 shocks to the car choice; solve discrete choice with continuous saving; convert annual to monthly; state utility curvature
Rust + Li et al.	Conflict under CI	A latent state known to the household violates CI; needs a Markov-unobservable estimator (pending)
Any structural likelihood + Heckman	Needs initial-state model	Required once household types are added
Adams et al. + structural model	Validation only	Static probit on subprime applicants; coefficients cannot be inserted
Rust/Schiraldi + Kasahara–Shimotsu types	Needs assumptions	Their baseline needs full-support transitions common to all types; a deterministic age transition violates it, so the propositions do not apply directly
Rust/Schiraldi + Higgins–Jochmans types	Compatible in form; weak in practice	No full-support requirement, but the rank condition needs types to differ in their initial-state distribution or in their response to a covariate; with rare purchases the signal is small
Type mixture + Keane–Wolpin initial-state device	Compatible under stated conditions	With type probabilities saturated in the first observed state, the likelihood equals Heckman's conditional likelihood (our reconstruction); needs entry into the data to be exogenous given type and state
Purchase-timing panel + Card–Hyslop device	Unsuitable	Their device needs a common initial state; selecting households to create one would condition on type-dependent outcomes
Rust-type replacement + Aguirregabiria–Gu–Luo	Limited	The replacement cost and type distribution are absorbed by the fixed effect; duration effects are identified only if flat beyond a threshold, and each unit needs two observed renewals
Normalized structural model + counterfactuals (Aguirregabiria–Suzuki; Arcidiacono–Miller)	Payoff changes only	Known additive payoff changes are identified under any normalization; changes in transitions or β are not unless the payoff of one action is known

Parameters do not transfer across period lengths or shock scales without conversion: an annual $\beta = 0.9$ is about 0.991 per month, and probit and logit coefficients differ in scale by a factor of roughly 1.6 to 1.8.

3 Broader Literature on Vehicle Purchase Timing

3.1 The Economic Question: Why Does a Household Purchase a Vehicle at a Particular Time?

The object of interest is not simply the number of cars demanded in a market or the type of car a consumer selects. It is the occurrence of a vehicle purchase over a specified future interval. At a forecasting date t_0, the empirical question is whether the information then available about a household helps explain the probability that it will acquire a vehicle over the following six or twelve months, with twenty-four months as an extension. A household that does not purchase during the next six months may nevertheless purchase during the subsequent six months; another may replace an existing car without changing the number of cars owned. These are not equivalent observations. A theory of automobile purchase timing must account for the household's current transportation position, its capacity to finance a transaction, the alternatives it considers, and the value of delaying the decision.

This section examines the evidence underlying the three-part framework introduced in Section 1: *Need*, *Ability*, and *Choice*. Need denotes the circumstances that create or alter the motivation to change the household's vehicle holdings. Ability denotes the financial opportunities and constraints that make particular transactions feasible. Choice denotes the selection among feasible actions, including purchasing a particular vehicle, retaining the existing fleet, relying on another transportation mode, or waiting. The layers organize the explanation of a purchase; they are not assumed to be separately observed, independent, or necessarily chronological. A change in employment may simultaneously alter travel requirements and income, while an anticipated purchase may lead a household to accumulate savings before a vehicle is actually needed. The literature informs the candidate formulation in the companion research proposal; it does not uniquely determine its functional forms or establish identification.

A distinction in the literature is essential at the outset. Aggregate differentiated-product demand models explain market shares, substitution patterns, prices, and product characteristics. Individual or household models of vehicle holding and transactions address whether, when, and how an existing fleet changes. Duration and event-history methods focus on the time until such a change occurs. These traditions overlap but do not estimate the same object. The choice-based foundations of McFadden (1974) and the aggregate-demand estimators of Berry (1994); Berry et al. (1995) are important to the vehicle-selection component; they are not, by themselves, household-level forecasts of purchase timing. The latter problem has a more direct empirical lineage in vehicle-holding duration and competing-risk transaction models (de Jong, 1996; Yamamoto et al., 1999; Mohammadian and Rashidi, 2007).

3.2 A Decomposition of Need, Ability, and Choice

The three terms identify different questions that are often combined in a single purchase indicator. Observing a purchase establishes that a transaction took place, not why it took place. Observing no purchase is even less informative: the household may be content with its current transportation, may want to purchase but lack acceptable financing, or may be capable of purchasing and prefer to wait. The distinction is therefore analytically consequential. It also prevents financial ability from

being confused with willingness to purchase and distinguishes a change in the attractiveness of a car from an increase in the need for transportation.

What belongs in the Need layer? Need should not be reduced to the arrival of a single exogenous shock. The relevant questions are whether the household's existing transportation arrangements continue to meet its requirements; which changes make them less satisfactory; how quickly the discrepancy is expected to grow; and whether the change is temporary, permanent, anticipated, or unexpected. There are several distinct pathways:

1. *Fleet deterioration and contractual deadlines.* An existing vehicle may become less reliable, costly to maintain, less valuable at trade-in, or due to be returned at lease expiration. The household can keep the vehicle, repair it, or begin a replacement process; deterioration raises pressure to adjust but does not uniquely determine when a purchase must occur.

2. *Household composition.* Childbirth, partnership formation or dissolution, the arrival or departure of another adult, and a teenager reaching driving age may change the number of drivers, the number of required trips, or the characteristics that a suitable vehicle must have.

3. *Employment and spatial circumstances.* A new job, a different commute, residential relocation, parking availability, and access to public transportation may alter the services a household obtains from its existing fleet and from non-car alternatives.

4. *Sudden losses and unexpected changes.* Theft, collision, or mechanical failure can rapidly reduce available transport capacity; conversely, the loss of a job or an adult leaving the household can reduce the desired stock of vehicles.

5. *Anticipated transitions and discretionary upgrading.* A household may replace a functioning car before an expected birth or move, or because a safer, more efficient, or otherwise preferred product becomes attractive. A strict rule requiring a discrete emergency or life event before every purchase would exclude these transactions by assumption.

These channels are not proposed as mutually exclusive event categories. Their purpose is to specify the economic content hidden by a generic indicator for "need to buy." A deterioration process, for example, may coincide with an employment change; a desire to upgrade may become urgent when the current vehicle requires a large repair.

What belongs in the Ability layer? The first question is not simply whether a bank would approve a loan. It is which purchase and financing arrangements are feasible under the household's available resources and prevailing terms. Relevant subquestions include the amount of immediately accessible cash; the affordability of an initial payment and recurring obligations; the consumer's access to unsecured or secured credit; the effect of interest rate, maturity, loan-to-value limits, and down-payment requirements; the equity or debt attached to a vehicle being traded; and whether a cheaper used car, a lease, or a cash purchase changes feasibility. These conditions may determine the *set* of attainable vehicles rather than provide a binary answer to whether any car can be purchased. They are also time-dependent: wealth, income, outstanding debt, credit supply, and quoted loan terms can change before the transaction occurs.

What belongs in the Choice layer? Even with purchasing motivation and adequate resources, the household faces multiple margins of adjustment. It may add a vehicle, replace one, retain the existing fleet, dispose of a car without replacement, or obtain transportation through another mode.

Conditional on acquiring a vehicle, it must select attributes such as vehicle class, vintage, technology, price, and financing arrangement. The household also chooses *when* to act. Buying now sacrifices the opportunity to wait for more information, a different price, or more favorable financing; waiting may incur repair costs, unreliable mobility, or lost convenience. A model that collapses all these possibilities into "buy/not buy" may still predict a binary outcome, but the omitted distinctions limit its behavioral interpretation.

Interaction rather than mandatory gates. The ordering Need–Ability–Choice is a way to reason about the transaction, not an assumption that a need event must first occur, financing must next be approved, and a choice must then be made. Some purchases occur without identifiable objective need; some households adapt their chosen vehicle to financing constraints; and some households anticipate a future need and prepare financially. For that reason, this review uses the three layers as potentially interdependent mechanisms. A sequential probability decomposition may be convenient for a carefully defined set of binary events, but no independent-filter interpretation is required by the framework. Whether each mechanism can be identified as a distinct statistical quantity is an empirical question addressed below.

3.3 Need: Durable-Goods Adjustment and Changing Transportation Requirements

3.3.1 Why replacement pressure can accumulate without an identifiable shock

Automobiles are durable assets that deliver transportation services over multiple periods. The household's existing vehicle stock can remain usable after its preferred stock or attributes have changed. Adjusting the stock involves transaction costs, search effort, possible losses in resale value, and the commitment of financial resources. These costs provide an economic explanation for periods of inaction punctuated by relatively infrequent transactions. In a stylized (S, s) adjustment rule, the household tolerates deviations from a preferred durable-goods position until an adjustment boundary is crossed. The threshold is endogenous to preferences, costs, income risk, and existing holdings; it is not the occurrence of a separately observed life event.

Eberly (1994) tests an (S, s) rule using automobile purchases and relates the distribution of durable-goods holdings to aggregate expenditure dynamics. The evidence is consistent with this type of adjustment for a substantial subset of households, but not for all households. The result is important because it provides a mechanism through which a need to adjust can accumulate gradually. Attanasio (2000) further studies estimation and aggregation of heterogeneous (S, s) rules for automobiles, emphasizing that inertia and timing can emerge from household-level adjustment behavior. Neither study implies that a household's latent need can be read directly from a single observable event. Rather, each motivates attention to the starting fleet, the desired fleet, the cost of adjustment, and heterogeneity in the adjustment threshold.

This literature raises several questions for the proposed Need layer. Does the household own no vehicle, one vehicle, or several vehicles? If it owns a car, how long has it held it, what is its age and condition, and what costs would continued ownership impose? Is the contemplated purchase a replacement, an addition, or an upgrade? A household with one unreliable vehicle and a household with two relatively new vehicles may experience the same income change yet face different pressures to transact. Conversely, two households with cars of the same age may differ because their maintenance costs, travel patterns, and willingness to tolerate failure differ. Age of the vehicle is therefore a potentially useful state variable, not a sufficient definition of need.

3.3.2 Household lifecycle events, initial ownership, and mobility context

The transportation literature supplies direct empirical evidence that vehicle transactions are associated with changes in household circumstances. Using differently structured data from France and Japan, Yamamoto (2008) studies vehicle transactions and changes in household ownership, finding associations with changes in adults, employment, and residence. The French analysis concerns transaction hazards, while the Japanese analysis concerns changes in vehicle counts. A notable qualification is that the improvement in statistical fit attributable to life-course event covariates is relatively modest in that study. An association with events therefore does not establish that such events alone explain a large proportion of transaction timing. Oakil et al. (2014) explicitly investigate whether car ownership changes occur in anticipation of major events or upon their occurrence, providing evidence that transactions can precede expected changes such as childbirth and employment transitions.

Initial ownership matters. Clark et al. (2016) distinguish movements from zero to one car, one to two cars, and reductions in the fleet using a large UK household panel. The effects of household composition and spatial context vary with the starting state: a life event need not have the same implication for a household without a car as for one already owning two. Public transport accessibility likewise changes the value of additional vehicle ownership. The framework therefore must not treat "need" as an invariant response to the same nominal event. Its meaning depends on current holdings, available alternatives, and the number and characteristics of household members.

More recent evidence reinforces both the significance and the limits of lifecycle factors. Gu et al. (2021) permit distinct household segments in a latent-class competing-risks formulation and report heterogeneous responses to life-course changes. Using the U.S. Panel Study of Income Dynamics, Oshanreh et al. (2025) examine vehicle purchase timing with demographic lifecycle covariates. Its use of interval-observed purchases and household-level records is germane to a forecast requiring precise monthly horizons: the existence of a statistically useful lifecycle variable does not establish that the purchase month can be identified from the dataset. Hossain and Fatmi (2025), in contrast, study the transition from zero-car status to a household's first vehicle and find that employment, residential, and family changes relate to how long a household remains car-free. That event is informative but differs from replacement of an existing car.

These studies justify a Need mechanism incorporating both persistent states and changing events. They also support a narrower conclusion than a universal "event triggers purchase" rule. Life events sometimes change the number of vehicles required, sometimes change vehicle characteristics without changing the fleet size, and sometimes create incentives to dispose of rather than acquire a car. A forecasting system must distinguish these outcomes if Need is to carry explanatory content.

3.3.3 Need as an evolving, potentially unobserved state

One conceptual representation is a difference between desired transportation services and those supplied by the current fleet and available alternatives. Call this difference a *mobility mismatch*. This is an organizing construct, not an observed variable or an already-estimated index. It may increase because a current car deteriorates, because household travel requirements increase, or because a new type of vehicle offers valued services. It may decrease after a move nearer to work or after an adult leaves the household. The relevant empirical questions are whether this mismatch can be proxied by observed ownership, trip-making, vehicle condition, lifecycle events, and spatial characteristics, and how much purchase-timing variation such proxies explain. This conception preserves both shock-driven purchases and purchases following gradual adjustment pressure without imposing a zero probability of discretionary acquisition.

3.4 Ability: Liquidity, Borrowing Constraints, and the Feasible Vehicle Set

3.4.1 Financial resources are not equivalent to current purchasing capacity

A household may wish to change vehicles yet lack sufficient cash to make the required down payment or to absorb the resulting monthly payment. Conversely, a household that would not qualify for a large loan may be able to purchase an inexpensive used car or transact without borrowing. These cases show why Ability should be represented as financial feasibility relative to a proposed transaction, not as a universal binary attribute of the household. Relevant resources include current income and its stability, liquid balances, available down-payment funds, net equity in the current vehicle, existing debt commitments, and the terms on which additional credit is offered. The constraints apply jointly to the vehicle selected and the financing used.

Evidence from auto finance illustrates the difference between wealth, income, and usable liquidity. Attanasio et al. (2008) estimate loan-demand responsiveness to interest rate and maturity using automobile loan contract data. Except for high-income households, the study finds substantial responsiveness to maturity and weaker responsiveness to the interest rate, a pattern the authors interpret as consistent with binding borrowing constraints. Longer maturity can make a vehicle easier to finance by reducing the required periodic payment, even though it need not reduce the total economic cost of borrowing. The distinction is relevant for a timing question: a financing offer may shift a prospective purchase into the current period without changing the household's underlying transportation requirements.

The study by Adams et al. (2009) examines a subprime auto retailer and finds that purchases respond strongly to the availability of cash during tax-rebate season and to down-payment requirements. The result is particularly revealing because a household may experience a temporary improvement in the capacity to transact without a commensurate improvement in permanent income. The study also emphasizes informational problems faced by lenders, so a household's access to credit need not be determined solely by its willingness to borrow. The study by Jiang et al. (2021), using a consumer panel with credit and employment information, reports that lump-sum bonus payments are associated with increased demand for auto loans, whereas the effects on subsequent delinquency differ from those of an increase in base salary. The distinction suggests that temporary funds and recurring earnings may enter household financing decisions differently. These findings support modeling the timing and form of available resources, not merely annual income or a static credit score.

3.4.2 Ability changes the vehicle choice set and the transaction itself

The financial constraint operates at several linked margins:

1. *Up-front funds.* What cash, trade-in equity, rebates, or other resources can cover a down payment, taxes, and transaction fees? Does negative equity on the existing vehicle reduce those resources?

2. *Recurring payment capacity.* What monthly obligation can the household sustain alongside housing, other debt, insurance, maintenance, and living costs? Does the prospective obligation remain affordable under a plausible income shock?

3. *Credit availability and contract terms.* Which lenders, maturities, rates, maximum principal amounts, and collateral restrictions are available? An available loan is not necessarily a loan the household finds attractive.

4. *Substitution across vehicles and financing modes.* Can a less expensive used vehicle, smaller down payment with a different lender, leasing, or outright cash purchase satisfy the transportation objective?

5. *Future financial conditions.* Would postponement permit more savings, repayment of existing debt, improvement in credit standing, or access to more favorable rates? Alternatively, could income uncertainty make postponement preferable despite current affordability?

The interaction between financing conditions and vehicle selection has direct empirical support. Argyle et al. (2021) exploit variation in credit terms across a large sample of used-car transactions. Borrowers offered shorter loan maturities respond by reducing expenditure, substantially through shifts to lower-quality vehicles, while some adjustment also occurs through negotiation of vehicle prices. In this setting, the consumer's effective choice set is not fixed independently of credit terms. This undermines a simplistic "approve financing, then independently choose a car" interpretation. Financing constraints, car quality, and transaction price are jointly relevant to what a household ultimately purchases.

Two further distinctions are needed. First, loan approval concerns a lender's decision as well as household affordability. An applicant can be declined or offered unattractive terms, while another household may finance the same vehicle with cash or a different lender. Second, the observed set of loan originations is selected: it excludes rejected applicants, potential purchasers who never apply, and purchases made without an observed loan. Consequently, one cannot infer the frequency of unmet Need or the degree of financial Ability from originated loans alone. The finance literature identifies mechanisms, but separate data and assumptions would be required to measure how these mechanisms operate in a general population of prospective vehicle buyers.

3.4.3 What the Ability literature establishes and leaves unresolved

Taken together, Attanasio et al. (2008); Adams et al. (2009); Argyle et al. (2021); Jiang et al. (2021) establish that borrowing terms and liquidity are economically consequential for vehicle transactions. Their outcomes, however, are not interchangeable. Some studies investigate loan demand or borrowing contracts, others observe vehicles purchased through a particular sales or financing channel, and still others study the car's price or quality. None, by itself, identifies the probability that a particular household will acquire *any* vehicle over a future six-month interval. The implication for the framework is twofold: financial feasibility must be allowed to affect both the likelihood and the characteristics of purchase, and the outcome under investigation must distinguish a vehicle purchase from the origination of a loan at a particular institution.

3.5 Choice: Available Actions, Substitution, and the Option to Wait

3.5.1 Vehicle purchase is a choice among heterogeneous alternatives

A household choosing whether to transact faces more than a homogeneous automobile product. A possible decision sequence is whether to change the current fleet, whether to add or replace a car, which vehicle type to obtain, and how to finance the transaction. The alternatives also include retaining and repairing existing vehicles, disposing of a car without replacement, public transportation, ride services, and other mobility arrangements. These possibilities cannot all be treated as identical forms of "not buying." The outside option in a car-choice model is a modeling category; its real economic content depends on the household's current holdings and mobility environment.

Random-utility models supply a foundation for representing substitution among differentiated alternatives. The conditional-logit formulation of McFadden (1974) makes the probabilities of alternative choices depend on their attributes and on the specified distribution of unobserved utility. Standard multinomial logit entails restrictions on substitution, most familiarly independence of irrelevant alternatives under its maintained error assumptions. Nested specifications relax some of these restrictions by permitting groups of alternatives to share unobserved components. Applying nested logit to the automobile holdings of one-car households, Berkovec and Rust (1985) demonstrate an early disaggregate approach to a large set of makes, models, and vintages. The substantive lesson is not that a particular nesting structure is automatically correct: it is that the definition of the feasible alternatives, the grouping of close substitutes, and the relevant ownership state matter to the resulting choice probabilities.

The aggregate demand contributions of Berry (1994); Berry et al. (1995) address a distinct problem. In markets where firms set prices in response to product characteristics observed by them but not by the analyst, price may be correlated with unobserved quality. Berry (1994) derives an inversion from market shares to mean product utility to facilitate instrumental-variables estimation; Berry et al. (1995) estimate differentiated-product automobile demand with heterogeneous tastes and an oligopoly supply side. This literature is essential for reasoning about substitution and price endogeneity in product markets. It does not directly recover the current liquidity, lifecycle state, or future purchase date of a named household from aggregate shares. The two levels of analysis should therefore be cited for their specific contributions rather than treated as interchangeable estimators.

3.5.2 The household's nested decisions do not necessarily form a fixed sequence

For purposes of the framework, the Choice layer can be decomposed into at least four related decisions:

1. *Adjustment versus continuation.* Does the household modify its fleet this period or continue with its current arrangement?

2. *Transaction type.* If it adjusts, does it acquire an additional vehicle, replace an existing one, or dispose of a vehicle without replacement?

3. *Vehicle attributes.* Conditional on acquisition, which class, age or vintage, technology, fuel economy, quality, and price tier provide the best match to preferences and constraints?

4. *Transaction and financing arrangement.* Should the household buy new or used, lease, pay cash, obtain a loan, use a trade-in, or postpone the entire transaction?

The listed sequence is a decomposition for analysis, not proof that consumers decide in that order. A monthly-payment ceiling can cause a household to change the desired vehicle type before making a final decision to buy; observing an attractive vehicle offer can accelerate the purchase date. The recent first-purchase study by Hossain and Fatmi (2025) connects a joint model of vehicle attributes to the hazard of leaving the car-free state through a logsum measuring the attractiveness of vehicle alternatives. The significance of that logsum in the timing equation provides evidence that the character of the available vehicle options and the timing of acquisition are related. It is not evidence that an identical specification will apply to households replacing cars they already own.

3.5.3 Waiting is an active economic alternative

Static choice models represent the alternatives available at a particular occasion but do not by themselves specify how the household values future opportunities. In a durable-goods market the

household generally retains the option to purchase later. Continuing with the current car preserves liquidity and permits future learning about income, prices, reliability, and product offerings; it can also impose inconvenience, repairs, or risk of failure. Hence "does not buy today" is not equivalent to the absence of preferences. It can be a positive selection of the continuation alternative.

Dynamic automobile models make this intertemporal margin explicit. Adda and Cooper (2000a) develop a stochastic dynamic model of household vehicle ownership to examine French scrappage subsidies. The analysis shows that incentives can bring purchases forward, affecting subsequent market activity through the distribution of vehicle ages. Schiraldi (2011) develops a structural dynamic model of new and used automobile replacement in which observed holding and buying behavior informs estimation of transaction costs. These papers offer theoretical support for treating waiting, holding, and purchasing as related options. They do not imply that every predictive application requires estimation of a full structural dynamic program.

The action-value equation of a dynamic discrete-choice model expresses this intertemporal comparison: a feasible action delivers current utility and a discounted expected continuation value (Rust's (4.1)–(4.4), Section 2.1.1). The state contains the current fleet, transportation requirements, financial circumstances, and relevant market information. Waiting can be preferred because its total expected value exceeds the value of immediate purchase, because the current fleet remains adequate, or because present financing terms make purchase unattractive. This is a decision-theoretic rationale, not an identification result. The data and assumptions needed to distinguish these explanations remain to be established.

3.5.4 Choice and transaction timing are related but different outcomes

A product-choice model may condition on the fact that a consumer has entered the market. A purchase-timing model must explain the time until market entry or transaction and distinguish households that are still at risk of purchasing from those whose event has already occurred. Similarly, conditioning on the vehicle actually bought creates selection: the available sample contains only purchasers and does not identify preferences among non-purchasers without additional assumptions. A framework that aims to predict automobile acquisition must therefore avoid treating observed buyers as a representative sample of all households potentially deciding whether to buy.

3.6 From Decisions to Time: Duration, Competing Risks, and Vehicle Transactions

3.6.1 Why transaction timing is not reducible to static ownership

Ownership at a particular date is a stock. Vehicle purchase, replacement, and disposal are flows that change or renew that stock. A cross-sectional indicator for owning one or two cars cannot establish when a car was purchased, whether it replaced an older car, or whether the household plans to transact soon. A household that recently replaced its only vehicle and a household that has held an unreliable vehicle for fifteen years can have identical fleet sizes but markedly different near-term adjustment pressures. The time dimension therefore requires information about the beginning and duration of the relevant ownership or non-ownership spell as well as the event that eventually ends it.

Duration analysis is concerned with an event occurring after time spent in a state. A hazard is the conditional probability or instantaneous rate of an event given that the state has survived until the relevant time. In economic applications, Kiefer (1988) provides a general treatment of duration data and hazards; the transport literature applies related ideas to vehicle holdings. de Jong (1996) models elapsed time between vehicle transactions jointly with vehicle type and use. The

study compares alternative duration assumptions and incorporates heterogeneity and time-varying covariates. This is an early recognition that the length of vehicle holding and the vehicle selected at replacement belong to a related system rather than to unrelated cross-sectional analyses.

3.6.2 Acquisition, replacement, and disposal are competing events

The distinction among transaction types predates the 2007 transaction study discussed below. Yamamoto et al. (1999) specify competing risks for replacing a vehicle, disposing of one without replacement, and acquiring an additional vehicle. These outcomes have distinct implications for the household fleet and potentially distinct determinants. Mohammadian and Rashidi (2007) extend the empirical vehicle-transaction literature by developing and comparing independent hazard models and a competing-hazards formulation for purchase, trade, and disposal. The authors include dynamic household and location characteristics, time-varying income and number of adults, and an externally supplied logsum for vehicle class and vintage. Their construction is especially relevant to the proposed framework: it recognizes that transaction timing responds to both the household's evolving circumstances and the desirability of available vehicles. It does not, however, estimate a separate latent Need equation, a financing-feasibility equation, and a conditional Choice equation.

The competing-risks framework avoids interpreting every exit from an ownership state as the same event. A vehicle disposal without replacement may reflect reduced transport need or financial pressure, whereas an acquisition may indicate an additional driver or changing preferences. Replacement can produce a vehicle purchase without increasing fleet size. If the observable endpoint is *any purchase*, an acquisition and a replacement generally both count as purchases; disposal without acquisition is a different transition. The exact mapping depends on the transaction codes available in a particular dataset, since labels such as "trade" and "replacement" may be recorded differently.

3.6.3 The relationship between hazards and purchase probabilities

To distinguish the endpoint used in much of this literature from the main purchase target, fix the end-of-month forecast origin t_0. Let $\widetilde{T}_h$ be the month of the household's *first transaction of any type* after t_0, and J_h its type. For mutually exclusive types $k \in \mathcal{K}$ and months ahead $r = 1, 2, \ldots$, define the forecast-origin cause hazard

$$\lambda^{\mathrm{tr}}_{hk}(r; t_0) = \mathbb{P}\Big(\widetilde{T}_h = t_0 + r, J_h = k \mid \widetilde{T}_h \geq t_0 + r, \mathcal{I}_{h,t_0}\Big). \tag{1}$$

These probabilities condition on the information available at t_0 and on remaining transaction-free. They integrate over later histories within that risk set, rather than substituting future observations. The probability of no transaction through month $t_0 + r$ is

$$S^{\mathrm{tr}}_h(r; t_0) = \prod_{u=1}^{r} \left\{ 1 - \sum_{k \in \mathcal{K}} \lambda^{\mathrm{tr}}_{hk}(u; t_0) \right\}, \qquad S^{\mathrm{tr}}_h(0; t_0) = 1. \tag{2}$$

The cumulative incidence of a first transaction of type k by horizon H is

$$F^{\mathrm{tr}}_{hk}(t_0, H) = \sum_{r=1}^{H} S^{\mathrm{tr}}_h(r-1; t_0) \lambda^{\mathrm{tr}}_{hk}(r; t_0). \tag{3}$$

If $\mathcal{P} \subseteq \mathcal{K}$ collects additions and replacements coded as purchases, then

$$\mathbb{P}\Big(\widetilde{T}_h \leq t_0 + H, J_h \in \mathcal{P} \mid \mathcal{I}_{h,t_0}\Big) = \sum_{k \in \mathcal{P}} F^{\mathrm{tr}}_{hk}(t_0, H). \tag{4}$$

Equation (4) is the probability that the *first transaction is a purchase* within the horizon. It differs from the probability that *any* purchase occurs within the horizon, the target in the companion proposal: a disposal followed by a purchase contributes to the latter but not to the former. Leasing is outside the purchase endpoint used in this draft and must be coded separately wherever observed.

These first-event identities clarify why a period hazard and a twelve-month probability are different quantities. They also show why a cause-specific hazard alone does not give the absolute probability of a particular first transaction: competing events alter the probability of remaining at risk. The distinction between a cause-specific hazard and cumulative incidence is fundamental in the competing-risks literature (Andersen et al., 2012). Models whose hazards instead condition on evolving covariates must integrate those histories consistently to obtain a forecast available at t_0.

Several observational issues follow directly. A household's recorded ownership spell may have begun before it enters the panel, creating left truncation or uncertainty about the elapsed duration; the last observation may precede the next transaction, creating right censoring. Surveys conducted annually or biennially may identify that a transaction occurred between interviews but not the exact month, producing interval censoring. A household may make more than one transaction over an extended horizon. In that case, a first-event competing-risks formulation does not by itself describe the complete recurrent purchase process; a multistate or recurrent-event extension would be necessary if the research objective requires subsequent acquisitions and replacements. These distinctions must be resolved before statistical estimation, but they do not alter the underlying Need–Ability–Choice explanation.

3.6.4 Model comparison and empirical extensions

The transaction-duration literature has evolved beyond independent hazards. Rashidi and Mohammadian (2011a) consider discrete observation intervals and unobserved heterogeneity in competing vehicle-transaction hazards. Rashidi and Mohammadian (2011b) formulate a dynamic hazard-based system connecting vehicle transactions, residential mobility, and job relocation decisions, rather than treating such changes as unrelated regressors. A related joint study by Rashidi et al. (2011) explicitly addresses interdependencies between vehicle transactions, relocation, and job changes. These studies provide methodological precedent for analyzing household decisions that co-evolve with the transportation state.

Hossein Rashidi and Mohammadian (2016) subsequently propose a discrete competing proportional-hazards model with a nested trivariate copula structure to represent dependence among unobserved components of different transaction types. Importantly, the correlation represented is among transaction outcomes, not direct evidence that latent Need, Ability, and Choice disturbances have the same structure. Gu et al. (2021) use a latent-class competing-risks model to capture heterogeneous household responses to life events. Heterogeneity and dependence solve different problems: a latent class permits different groups to have different behavioral parameters, whereas correlated disturbances express statistical association not explained by the specified observed variables. Both can be relevant, but neither automatically identifies the three proposed mechanisms.

Another comparison is between duration and discrete-choice approaches. Ghasri et al. (2018) examine conceptual and practical differences in modeling time-dependent vehicle ownership outcomes, including differences in data construction and the interpretation of the outcome. Hossain and Fatmi (2025) explicitly join a duration model for first purchase to a vehicle-type choice model. These contributions imply that transaction timing, alternative attractiveness, and heterogeneity can be handled through several econometric architectures. They do not establish that a particular architecture is uniformly preferable across available datasets and forecasting tasks.

3.6.5 The adjacent literature on customer-level purchase timing

Marketing research supplies a complementary perspective because it asks when a household is likely to make its next purchase, sometimes with incomplete purchase histories. In a conditional hazard model of household purchasing, Jain and Vilcassim (1991) examine interpurchase times as functions of observed characteristics, marketing variables, and unobserved heterogeneity. Vakratsas and Bass (2002) use a finite-mixture accelerated failure-time framework to represent segments with different purchasing rates and responses to marketing activity. These are not vehicle-transaction studies and should not be cited as direct evidence for automobile replacement behavior. Their relevance lies in the construction of prospective purchase probabilities and the explicit recognition that the same observable characteristics may conceal heterogeneous purchase rates.

Rossi et al. (1996) demonstrate that household purchase histories can have business value for targeting. For the present banking application, that finding motivates a further distinction: predictive rank ordering, the incremental effect of an offer, and the profitability of acting on a forecast are different outcomes. A household likely to buy a car soon may not be induced to borrow by receiving a promotion, and a household likely to borrow may not select a particular lender. Thus the vehicle-purchase forecast should first be defined independently of a proposed intervention, while any claim about the effectiveness of contacting the household would require additional evidence. The marketing literature motivates attention to household-level histories and heterogeneous purchase rates without replacing the economic content of Need, Ability, and Choice.

3.7 Dependence Across Layers: Economic Mechanisms, Probability, and Identification

3.7.1 Why the three mechanisms can move together

Need, Ability, and Choice can be empirically associated even when they answer distinct economic questions. Consider a household expecting a child. The anticipated change may increase demand for a larger vehicle, while the household simultaneously accumulates a down payment and searches for a suitable car. An observer who sees the eventual purchase cannot infer whether a financing change or a shift in mobility requirements was decisive. Similarly, employment relocation can change both commuting needs and income. A borrower may prefer a more expensive vehicle and attempt to stretch the financing arrangement; a restrictive maturity offer may instead cause selection of a cheaper vehicle. The joint-decision and financing literature supplies credible examples of these interactions (Oakil et al., 2014; Rashidi and Mohammadian, 2011b; Argyle et al., 2021).

The examples motivate a conceptual system in which the propensity to adjust the vehicle stock, financial feasibility, and utility from available actions depend on overlapping observed and unobserved household characteristics. Let N_{hr} represent a possibly latent state of mobility or replacement pressure, and let A_{hr} summarize financial conditions and attainable options. The Choice mechanism maps these states, preferences, and market information into the probability of purchase. Schematically, one could write

$$\mathbb{P}(B_{hr} = 1 \mid \mathcal{I}_{hr}^{-}) = \int \mathbb{P}(B_{hr} = 1 \mid \nu, \alpha, \mathcal{I}_{hr}^{-}) \, dF_{N,A|\mathcal{I}_{hr}^{-}}(\nu, \alpha), \tag{5}$$

where B_{hr} indicates a purchase in decision month r and $\mathcal{I}_{hr}^{-}$ denotes the information available immediately before that decision. Here ν and α are possible values of the Need and Ability constructs, distinct from observed fleet size and vehicle age. The expression permits dependence between Need and Ability through their joint distribution, while the conditional purchase term represents the Choice outcome. It is only a generic probability representation, not an identification argument or a

specification selected for estimation. The choice of variables and the interpretation of the latent states remain to be established.

If instead all three layers were defined as required binary conditions N, A, and C, the exact probability chain rule would be

$$\mathbb{P}(N, A, C \mid X) = \mathbb{P}(N \mid X)\mathbb{P}(A \mid N, X)\mathbb{P}(C \mid N, A, X). \tag{6}$$

This expression does *not* assume independence. Multiplying the three *marginal* probabilities $\mathbb{P}(N \mid X)\mathbb{P}(A \mid X)\mathbb{P}(C \mid X)$, by contrast, would generally be unjustified under dependence. More fundamentally, defining a vehicle purchase as requiring an objectively identified Need event would impose a zero purchase probability in its absence, excluding discretionary acquisitions. Accordingly, the proposed framework retains the three economic mechanisms without asserting that they are universally necessary Boolean gates.

3.7.2 Joint estimation does not automatically establish identification

A statistical model may permit correlated residuals, common random effects, latent classes, or a copula dependence structure, but the ability to fit such a model is not the same as the ability to assign a unique interpretation to its components. Observed non-purchase can be consistent with weak motivation, binding financing constraints, or a high value of waiting. If these internal states are not observed, different combinations of structural parameters may generate the same distribution of recorded vehicle transactions. Repeated household observations can reveal histories and within-household variation, but do not, by themselves, identify every latent effect or the direction of causality. Credible separation may require direct measurements of financing offers and applications, information about existing fleets and life events, timing restrictions, credible excluded variables, or explicitly defended functional-form assumptions.

The precedent in Hossein Rashidi and Mohammadian (2016) is therefore methodological rather than a direct validation of a three-equation model of Need, Ability, and Choice. Copulas can represent dependence among specified disturbances; they cannot make unmeasured mechanisms separately observable. Similarly, the heterogeneous responses found by Gu et al. (2021) do not demonstrate that the underlying segments correspond to distinct Need or Ability types. A defensible literature review should distinguish a description of correlations, a behavioral interpretation of those correlations, and causal identification of an effect. This distinction is especially relevant when financial feasibility may be affected by the vehicle selected and by lender decisions made jointly with consumer applications.

3.7.3 Prediction does not require identifying every mechanism

The motivation for an economically decomposed framework is explanatory clarity and the ability to ask targeted empirical questions. Its contribution to prediction must nevertheless be evaluated separately. A reduced-form event-history model could predict purchases using vehicle age, household changes, and financial characteristics without uniquely identifying latent Need, Ability, and Choice. A more elaborate structural model might explain counterfactual responses to changes in financing terms, but it could impose assumptions that are difficult to validate. The appropriate comparison is not simply between "independent" and "joint" models; it is between models with demonstrably different information requirements, identifying assumptions, and predictive or counterfactual targets. The literature therefore supports investigating interdependence, while leaving open whether estimating separate correlated latent equations is feasible or necessary for the stated prediction objective.

3.8 Synthesis: What the Literature Justifies and What Remains to Be Shown

The literature provides distinct but complementary foundations for the Need–Ability–Choice framework. Durable-goods adjustment theory (Eberly, 1994; Attanasio, 2000) explains why a household can remain inactive while adjustment pressure accumulates; lifecycle and transportation studies (Yamamoto, 2008; Oakil et al., 2014; Clark et al., 2016; Gu et al., 2021) show that changing household circumstances and initial ownership states matter to actual transactions. The consumer-credit literature (Attanasio et al., 2008; Adams et al., 2009; Argyle et al., 2021; Jiang et al., 2021) demonstrates that liquidity, loan structure, and credit terms can affect the feasibility, timing, and composition of vehicle purchases. Discrete and dynamic choice models (McFadden, 1974; Berkovec and Rust, 1985; Adda and Cooper, 2000a; Schiraldi, 2011) establish why the household compares differentiated vehicles, current ownership, and waiting. Transaction-duration models (Yamamoto et al., 1999; de Jong, 1996; Mohammadian and Rashidi, 2007; Rashidi and Mohammadian, 2011a; Hossein Rashidi and Mohammadian, 2016) provide the apparatus for distinguishing the timing and types of observed fleet changes.

The decomposition is therefore theoretically motivated, but it is not yet an empirically verified three-layer structural model. The literature justifies keeping separate the questions *what creates the motivation to adjust?*, *which adjustments are financially possible?*, and *which feasible action is preferred at which time?* It also demonstrates why these questions cannot be answered by observing purchases alone. At least five unresolved issues follow for a research program organized around the framework.

First, the unit and event definition. A household-level acquisition is not the same as an individual's purchase or a loan originated at a particular bank. Fleet additions and vehicle replacements are different transactions; both may produce loan demand, but not necessarily with the same financing mechanism. The working estimand in the companion proposal specifies purchases through additions or replacements. Its implementation still requires transaction codes and a household definition that handle multiple vehicles and drivers consistently.

Second, the observability of Need. Fleet age, maintenance costs, lease expiration, lifecycle events, and transport accessibility may proxy for changing adjustment pressure, but none uniquely measures motivation. Research is needed to determine which of these variables are observable before the forecast date and whether their predictive value depends on the household's initial ownership state.

Third, the observability of Ability. Income and credit scores are incomplete measures of financial feasibility. Relevant information may include liquid assets, payment burdens, contract offers, down-payment requirements, and vehicle equity. Observed loan outcomes are themselves selected by applications and lender decisions; the implications for population-wide vehicle-purchase prediction need to be made explicit.

Fourth, timing, anticipation, and dependence. Household changes can affect several layers before the observed transaction. Published studies offer ways to model time-varying characteristics, endogeneity, unobserved heterogeneity, and dependence, but do not identify a unique three-layer specification. The data must support temporal ordering and distinguish historical information from future events that were not known at the forecasting origin.

Fifth, what constitutes a useful forecast. A time-to-event model produces period hazards, while the stated application requires cumulative probabilities of acquisition over multiple horizons. These probabilities should be coherent across horizons and calibrated against observed outcomes, with appropriate treatment of censoring and competing events. Comparisons with simpler survival or discrete-time models are required before attributing value to additional structure. This is a proposed evaluation criterion, not a claim of an established forecasting advantage.

Taken together, the reviewed work motivates treating automobile purchase timing as the observed result of evolving transportation requirements, financially feasible alternatives, and intertemporal choices. The purpose of the Need–Ability–Choice framework is to make those determinants explicit and expose the questions hidden inside a binary purchase indicator. The candidate utility formulation in the companion proposal makes these questions mathematically explicit. The next task is to establish which mechanisms can be measured and distinguished with available data, justify the associated assumptions, and evaluate the resulting purchase probabilities without using information unavailable at the prediction date.

4 Further References Identified for Later Reading

A bounded search for work not covered above turned up the references below. Each bibliographic entry was checked against the publisher's record (Crossref) or the source page. **These papers have not yet been read in full.** The one-line descriptions come from abstracts and front matter and record why each paper may matter for the purchase-timing model. They are not findings of this review, and nothing in Sections 2–3 relies on them. They are grouped by the model component they would inform, following the C1–C5 labels of Section 2.

Selected for full review next

These five are next in line for the same equation-and-assumption treatment as Section 2.

- Green et al. (2020): the Cash-for-Clunkers rebate as down-payment liquidity rather than credit capacity. It bears on how liquidity enters the purchase decision (C4).
- Baugh et al. (2021): even highly liquid households spend out of tax refunds. A test of whether an account balance is the right liquidity state (C4).
- Adda and Cooper (2000b): a household keep-or-replace model of car purchases with age and income shocks, aggregated to sales (C1).
- Keane and Wolpin (2007): validating a structural model on a holdout sample that faces a different policy regime (evaluation).
- Haider et al. (2020): calibration of predicted survival distributions at a fixed horizon (evaluation).

C1. Dynamic choice of when to buy: durable and vehicle replacement

- Attanasio et al. (2022): a life-cycle model separating whether to buy a car from how much to spend; purchases respond to expected income and its variability.
- Gillingham et al. (2018): a finite-horizon vehicle ownership, type, and usage model on Danish registers, with an explicit scrap-or-sell choice.

- Berger and Vavra (2015): a durable adjustment hazard estimated on household panel data; a reduced-form counterpart to the purchase probability.
- Bajari et al. (2013): a household-panel template for nonconvex transaction costs with durable, liquid-asset, and income states (housing).
- Gavazza et al. (2014), Chen et al. (2013), Stolyarov (2002): persistent differences in how much households value newer cars, lumpy transaction costs, and resale behavior that is not monotone in age. Together they bear on the latent-type interpretation and on the shape of the age effect.
- Copeland (2014): buyers shift purchase timing within the model year in response to prices; seasonality at a six-month horizon.
- Jacobsen and van Benthem (2015): the elasticity of scrappage with respect to used-car value.
- Glerum et al. (2013), Wakamori (2011), Archsmith et al. (2020): adding versus replacing a car in multi-vehicle households.
- Liu and Cirillo (2018), Bierlaire et al. (2021): dynamic choice models of vehicle purchase timing in transportation research.

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C4. Liquidity and purchase timing

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